Typshade Documentation

Version 0.1.4

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Abstract

Typshade is a Typst-native package for multiple-sequence alignment figures. It reimplements the practical feature set of TeXshade for shading and labeling nucleotide and peptide alignments, while exposing the interface as Typst-native building blocks: purpose-level recipes, named arguments, command helper lists, and reusable helper functions.

Contents

1 Design Goals

- Preserve TeXshade-style bioinformatics capabilities: shading, consensus, numbering, rulers, regions, features, graphs, sequence logos, subfamily logos, and structure tracks.
- Prefer Typst-native APIs over TeX macro compatibility.
- Make the common workflow short and readable through `shade(..., figure: (...))`.
- Keep low-level command helpers for exact, reproducible control.
- Avoid hidden global state. A figure should mostly describe itself where it is rendered.

2 Package Overview

Typshade is not an aligner. Like TeXshade, it assumes that a scientific alignment program has already produced an aligned nucleotide or peptide sequence file. Typshade then turns that alignment into publication-quality Typst content: colored residue cells, labels, rulers, consensus rows, logos, feature tracks, graphs, legends, and optional structure/topology annotations.

The package is organized around one rendering entry point, `shade`, plus a set of declarative helpers. The most important distinction from TeXshade is that state is represented by Typst values rather than by TeX macro side effects. This has practical consequences:

- Figure settings can be stored in variables, passed to functions, imported from other Typst files, and reviewed as data.
- The order of low-level commands still matters where the biology requires it, but purpose-level recipes reduce order sensitivity for common figures.
- Inspection helpers return Typst content/data instead of writing to the TeX log.
- Side-effect features such as exporting Chimera/Pymol command files are intentionally separated from document rendering.

2.1 Package Contents

Component	Purpose
`package/lib.typ`	Public package entry point. Import this file locally or as `@preview/typshade`.
`internal/interface`	Public-facing API modules: rendering, recipes, tracks, annotations, controls, inspection, and presets.
`internal/model`	Parsing, palettes, logo calculations, PDB selection helpers, and text-style helpers.
`internal/render`	Alignment table rendering, feature rows, graph tracks, and structure rows.
`docs/documentation.typ`	This manual and TeXshade-to-Typshade correspondence reference.
`examples`	Small runnable examples for package development and regression checks.

2.2 System Requirements

Typshade is a Typst package. It requires a Typst compiler compatible with the package metadata in `typst.toml`. It does not require LaTeX, `color.sty`, PostScript drivers, `pstricks`, or TeX memory tuning.

The package can read files that Typst is allowed to access under the active `--root`. When compiling local examples, make sure the project root includes the alignment and annotation files:

```
typst compile --root /path/to/project doc.typ out.pdf
```

3 Alignment Input Files

Typshade follows the TeXshade model: input files contain already-aligned sequences. It does not compute the alignment. Use external tools such as Clustal, MUSCLE, MAFFT, T-Coffee, or a domain-specific pipeline to create the alignment first, then read the result in your document and pass the bytes to `shade`. `read(..., encoding: none)` remains supported on all supported Typst versions, including Typst 0.15 and later. With Typst 0.15 or later, you may also pass a resolved `path(...)` to Typshade and let the package read the file. This keeps path resolution on the project side while preserving existing `read(..., encoding: none)` documents.

The `format:` argument defaults to `auto`, but explicit formats are recommended when using `read(..., encoding: none)` or `path(...)` for reproducible figures:

```
// Works on all supported Typst versions, including Typst 0.15+.
#let msf = read("alignment.msf", encoding: none)
#let aln = read("alignment.aln", encoding: none)
#let fasta = read("alignment.fasta", encoding: none)

#shade(msf, format: "msf")
#shade(aln, format: "aln", seq-type: "P")
#shade(fasta, format: "fasta", seq-type: "N")

// Typst 0.15+: path is resolved where it is constructed.
#shade(path("alignment.msf"), format: "msf")
```

Typeset result Input formats rendered with bundled fixtures

MSF

```
AQP1.PRO TLGLLLSCQISI 91
AQP2.PRO TVACLVGCHVSF 83
AQP3.PRO TFAMCFLAREPW 91
AQP4.PRO TVAMVCTRKISI 112
AQP5.PRO TLALLIGNQISL 84
```

ALN

```
      123
      |||
Alpha AEF. 3
Beta  ADF. 3
Gamma A.G. 2
```

FASTA


```
dna1 ATGC 4
dna2 ATGT 4
```

Typshade expects alignment inputs to be rectangular: every parsed sequence must have the same aligned column length. Empty inputs, mismatched sequence lengths, unknown explicit `format:` values, unknown sequence names, and out-of-range sequence indices are rejected with Typshade-level diagnostics before rendering. Sequence references are 1-based when numeric, or exact sequence names when textual.

3.1 MSF Files

MSF is the richest supported input format because it can carry sequence names, sequence type, lengths, and header metadata. Typshade uses the sequence names and aligned sequence blocks. If the MSF header declares peptide/nucleotide type, `seq-type: auto` can usually infer it.

Typical use:

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  figure: publication(region: "80..112"),
)

// Typst 0.15+ alternative:
#shade(
  path("AQPpro.MSF"),
  format: "msf",
  figure: publication(region: "80..112"),
)
```

Typeset result MSF input with publication recipe

```
          80        90        100       110
          |         |         |         |
AQP1.PRO  TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO  TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO  TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO  TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO  TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
consensus ! * *    ** *****!  !! *!!* ***!*
```

3.2 ALN Files

ALN/Clustal-like files often lack an explicit sequence type. In such cases, pass `seq-type: "P"` for peptide alignments or `seq-type: "N"` for nucleotide alignments. Minimal ALN files are acceptable as long as sequence names and aligned blocks are recoverable.

```
#shade(
  "AQP2spec.ALN",
  seq-type: "P",
  commands: (
    diverse(),
    window(1, "77..109"),
    ruler("top", sequence: 1),
  ),
)
```

Typeset result ALN input with diverse shading

```
      1 2 3
      | | |
Alpha . . . . 3
Beta  . . . . 3
Gamma . . g . 2
```

3.3 FASTA Files

FASTA files are supported for aligned sequences. Each record begins with `>`, and the first word is used as the default sequence name. FASTA does not encode alignment type, so explicit `seq-type:` is recommended for reproducible output.

```
#shade(
  "alignment.fasta",
  seq-type: "N",
  figure: logo-analysis(colors: "nucleotide"),
)
```

Typeset result FASTA nucleotide logo


```
dna1 ATGC 4
dna2 ATGT 4
dna3 ATGA 4
```

3.4 Supplemental Files

Some features need supplemental files:

File Kind	Used By	Notes
T-Coffee score data	`tcoffee(source)`, `tcoffee-scores(source)`	The score data must correspond to the displayed alignment. Pass `read(..., encoding: none)`.
PHD topology/ secondary files	`phd-topology-track`, `phd-secondary-track`, `structure-map`	Rendered as structure/feature tracks.
HMMTOP files	`hmmtop-track`, `structures`	Optional `hmmtop-sequence` selects a prediction within multi-entry files.
DSSP/STRIDE files	`dssp-track`, `stride-track`	Supports structure type filtering and appearance controls.
PDB files	`pdb-point`, `pdb-line`, `pdb-plane`, `pdb-selection`	Used for distance/plane-based residue selections.
Graph data sources	`graph(..., metric: read(..., encoding: none))`	Data should be normalized or have explicit ranges when needed.

4 Processing Model

TeXshade processes an environment by reading an alignment, resetting defaults, loading an optional parameter file, executing in-environment commands, then setting the alignment. Typshade keeps the same conceptual stages but expresses them as Typst values.

Step	TeXshade Model	Typshade Model
1	Analyze alignment file.	`shade(read(..., encoding: none), format: ...)` parses caller-supplied alignment bytes. Recipes may inspect the alignment before rendering.
2	Reset defaults.	The renderer starts from built-in defaults for every figure call.
3	Load parameter file.	`preset:`, `theme:`, imported Typst variables, or command arrays provide reusable settings.
4	Execute environment commands.	`figure:`, top-level arguments, `regions:`, `features:`, and `commands:` are merged explicitly.
5	Set alignment line by line.	The renderer builds Typst tables/blocks with deterministic layout.

The effective command order is:

1. `figure:` recipes or figure command fragments.
2. `preset:` and `theme:`.
3. Top-level convenience arguments such as `seq-type:`, `mode:`, `ruler:`, and `logo:`.
4. `regions:` and `features:`.
5. Explicit `commands:`.

This order lets recipes provide useful defaults while still allowing final manual overrides in `commands:`.

5 Quick Start

```
#import "@preview/typshade:0.1.4": *

#shade(
  "alignment.msf",
  theme: "screen",
  figure: motif-map(auto),
)
```

Typeset result Quick start motif map

```
           80        90        100       110
           |         |         |         |
AQP1.PRO   TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO   TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO   TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO   TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO   TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
consensus  ! * *    ** *****!  !! *!!* ***!*
```

6 Why The UI Is Different

TeXshade is configured by issuing many commands inside an environment. That is natural in TeX, but it makes large figures difficult to scan because related settings are spread across an order-sensitive command stream.

TeXshade style:

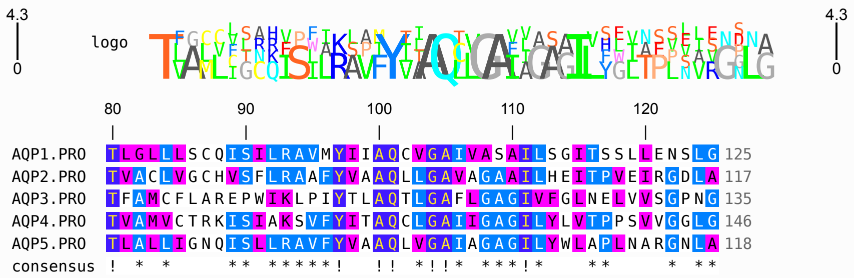
```
\begin{texshade}{alignment.msf}
  \shadingmode[similar]{identical}
  \shadingcolors{blues}
  \threshold{45}
  \residuesperline{45}
  \setends{1}{80..125}
  \showruler{top}{1}
  \rulersteps{10}
```

```
  \showconsensus{bottom}
  \showsequencelogo{top}
  \shaderegion{1}{NPA}{White}{BrickRed}
  \feature{top}{1}{NXX[ST]N}{box[Yellow]}{motif}
\end{texshade}
```

Typshade style:

```
#shade(
  "alignment.msf",
  figure: publication(
    similarity: "blues",
    region: "80..125",
    logo: "charge",
    motifs: (
      "NPA": (bg: "BrickRed", text: "active site"),
      "NXX[ST]N": "motif",
    ),
  ),
)
```

Typeset result Typshade publication recipe output



The Typshade form names the figure's scientific intent directly. This makes documents easier to scan, review, reuse, and parameterize.

7 Feature Overview

This section mirrors the role of TeXshade's package overview: it introduces the main capabilities before the command reference.

7.1 Predefined Shading Modes

Mode	Typshade API	Use Case
Identity	`identical(...)` or `scoring-mode("identical")`	Shade columns where a residue reaches the threshold. Useful for basic conservation figures.
Similarity	`similar(...)` or `scoring-mode("similar")`	Shade identical and chemically similar residues. This is the usual protein-alignment default.
T-Coffee	`tcoffee(read("scores.asc", encoding: none))`	Use an external T-Coffee score file to drive shading and graph tracks.
Diverse	`diverse(...)`	Highlight deviations from a reference sequence. Useful for variants and closely related homologs.
Functional	`functional("charge")`, `functional("hydropathy")`, etc.	Shade residues by biochemical property rather than only conservation.
Single sequence	`single-sequence(sequence: 1)`	Format one sequence with numbering, labels, translation/complement features, and manual regions.

Functional modes include `charge`, `hydropathy`, `structure`, `chemical`, `rasmol`, `standard area`, `accessible area`, and DNA-oriented functional shading. Custom functional group definitions use `clear-functional-groups` and `functional-group`.

7.2 Graphs And Color Scales

Feature rows can carry numerical information as bars, color scales, stacked bars, and conservation-derived plots. In TeXshade this is done through the general `\feature` command. In Typshade, the preferred high-level helper is `graph`:

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    window(1, "138..170"),
    graph("top", 1, "138..170", "conservation", kind: "bar"),
    graph("bottom", 1, "138..170", "hydrophobicity", kind: "color", options: ("GreenRed",)),
  ),
)
```

Typeset result Graph tracks and color scales

```
AQP1.PRO GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
```

Built-in graph metrics include `conservation`, `hydrophobicity`, `molweight`, and `charge`. Graph appearance can be tuned with `bar-graph-stretch` and `color-scale-stretch`.

7.3 Secondary Structures

Typshade can overlay structure information from DSSP, STRIDE, PHD, and HMMTOP files. For ordinary figures use `structure-map` or `structures`; for exact control use the format-specific track helpers.

```
#let alignment = read("AQPpro.MSF", encoding: none)
#let topology = read("AQP1.top", encoding: none)
#let secondary = read("AQP1.phd", encoding: none)

#shade(
  alignment,
  format: "msf",
  figure: structure-map(
    1,
    topology: topology,
    secondary: secondary,
  ),
)
```

Typeset result Structure map output

```
                     10        20        30        40
                      |         |         |         |
AQP1.PRO      1 MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLE 40
AQP2.PRO      1 MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...S 36
AQP3.PRO      1 M.........LLRQALAECLGTLILVMFGCGSVAQVVLSRG 44
AQP4.PRO      1 MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...G 61
AQP5.PRO      1 MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...S 37
consensus       !  *    ** *** !!*!*****!*** !!** *

                     50        60        70        80
                      |         |         |         |
AQP1.PRO     41 RNQTLVQDNVKVSLAFGLSIATLAQSVGHISGAHSNPAVT 80
AQP2.PRO     37 S....PPSVLQIAVAFGLGIGILVQALGHVSGAHINPAVT 72
AQP3.PRO     45 TH....GGFLTINLAFGFAVTLAILVAGQVSGAHLNPAVT 80
AQP4.PRO     62 SENPLPVDMVLISLCFGLSIATMVQCFGHISGGHINPAVT 101
AQP5.PRO     38 A....LPTILQISIAFGLAIGTLAQALGPVSGGHINPAIT 73
consensus                *  ****!!* * ** *  !**!!*!*!!!*!
```

Structure visibility and appearance are controlled by `show-structure-types`, `hide-structure-types`, `structure-appearance`, `use-first-dssp-column`, and `use-second-dssp-column`.

7.4 Fingerprints

The `fingerprint` control gives a compact overview of long alignments by reducing residue cells to thin visual marks. It is compatible with the same shading modes, feature rows, and legends as normal alignment figures.

```
#shade(
  "AQPpro.MSF",
  commands: (
    similar(colors: "grays", all-match-threshold: 100),
    fingerprint(360),
    legend(),
  ),
)
```

Typeset result Fingerprint overview

```
all match
gap
conserved
similar

AQP1.PRO MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLERNQTL
AQP2.PRO MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...SS....
AQP3.PRO M.........LLRQALAECLGTLILVMFGCGSVAQVVLSRGTH...
AQP4.PRO MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...GSENPL
AQP5.PRO MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...SA....
```

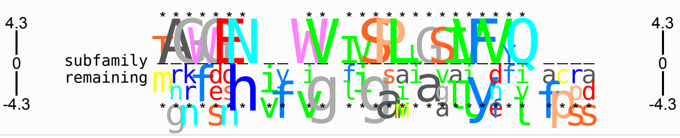
7.5 Sequence Logos And Subfamily Logos

Sequence logos show residue frequency and information content per alignment position. Subfamily logos visualize residues that distinguish a subset of sequences from the rest. Typshade supports both through explicit tracks and through `logo-analysis`.

```
#shade(
  "AQPpro.MSF",
  figure: logo-analysis(
    region: "203..235",
    colors: "charge",
    subfamily: (3,),
    relevance: (threshold: 1.0, char: "*"),
  ),
)
```

Typeset result Sequence and subfamily logos


```
AQP1.PRO  TRNFSNHWIFWVGPFIGSALAVLIYDFILAPRS 235
AQP2.PRO  TGKFDDHWVFWIGPLVGAIIGSLLYNYLLFPSA 227
AQP3.PRO  TAGWGENW..WWVPIVSPLLGSIVFVYQL.... 256
AQP4.PRO  MGNWENHWIYWVGPIIGAVLAGALYEYV.FCPD 255
AQP5.PRO  MNRFSSHWVFWVGPIVGAMLAAILYFYLLFPSS 229
consensus *  *  *! *!**!**** **  ** * ***
```



Frequency correction, logo colors, logo scales, negative subfamily values, and relevance markers are available through `frequency-correction`, `logo-color`, `logo-scale`, `negative-logo-values`, and `relevance-marker`.

7.6 Single-Sequence Figures

Single-sequence display is useful for formatted ORFs, UTRs, translated regions, complements, and motif annotations. The sequence can come from a one-sequence file or from one row of a larger alignment.

```
#shade(
  "AQPDNA.MSF",
  seq-type: "N",
  commands: (
    single-sequence(sequence: 1),
    shift-single-sequence(),
    window(1, "-84..525", start: -84),
    lower(1, "-84..-1,439..525"),
    mark("bottom", 1, "390..402", fill: "Blue", text: "poly-Arg"),
  ),
)
```

Typeset result Single-sequence translation/complement marks

```
AQP1nuc.SEQ  c c t g g g C A T T G A G A T C A T T G G C A C C C T G C A 443
```

7.7 Customization Surface

Most TeXshade customization categories have direct Typshade equivalents: sequence labels, numbering, rulers, gaps, regional highlighting, feature tracks, legends, captions, typography, and spacing. The important design change is that styling is local, inspectable, and composable:

```
#let paper-style = (
  typography(target: "all", size: 8pt),
  names(color: "RoyalBlue"),
  numbers(position: "right"),
  ruler("top", every: 10),
)

#let alignment = read("alignment.msf", encoding: none)

#shade(alignment, format: "msf", commands: paper-style)
```

Typeset result Reusable command style output

```
              80                  90                  100                 110
              |                   |                   |                   |
AQP1.PRO  T L G L L L S C Q I S I L R A V M Y I I A Q C V G A I V A S A I L 112
AQP2.PRO  T V A C L V G C H V S F L R A A F Y V A A Q L L G A V A G A A I L 104
AQP3.PRO  T F A M C F L A R E P W I K L P I Y T L A Q T L G A F L G A G I V 112
AQP4.PRO  T V A M V C T R K I S I A K S V F Y I T A Q C L G A I I G A G I L 133
AQP5.PRO  T L A L L I G N Q I S L L R A V F Y V A A Q L V G A I A G A G I L 105
```

8 Worked Examples

The examples below are intentionally small. They are meant to show how common TeXshade-style figures translate into idiomatic Typshade source.

8.1 Identity Shading With A Ruler

Use identity shading when exact residue conservation is the focus.

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    identical(colors: "blues", threshold: 50, all-match-threshold: 80),
    window(1, "80..112"),
    ruler("top", sequence: 1, every: 10),
    no-consensus(),
```

```
  ),
)
```

Typeset result

```
         80        90        100       110
         |         |         |         |
AQP1.PRO TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
```

The same figure can be expressed as a recipe when the intent is a standard publication panel:

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  figure: publication(
    mode: "identical",
    threshold: 50,
    region: "80..112",
    conservation: false,
  ),
)
```

Typeset result

```
           80        90        100       110
           |         |         |         |
AQP1.PRO   TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO   TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO   TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO   TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO   TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
consensus  ! * *    ** *****!  !! *!!* ***!*
```

8.2 Similarity Shading With Motif Labels

Similarity shading is the default for many peptide alignments because it can highlight chemically similar residues even when exact identity is low.

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    similar(colors: "blues", threshold: 45),
    window(1, "80..125"),
    ruler("top", sequence: 1, every: 10),
    consensus("bottom", name: "conservation"),
    motif(1, "NPA", text: "NPA loop", fg: "White", bg: "BrickRed"),
    motif(1, "NXX[ST]N", text: "glycosylation"),
  ),
)
```

Typeset result

```
             80        90        100       110       120
             |         |         |         |         |
AQP1.PRO     TLGLLLSCQISILRAVMYIIAQCVGAIVASAILSGITSSLLENSLG 125
AQP2.PRO     TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAILHEITPVEIRGDLA 117
AQP3.PRO     TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIVFGLNELVVSGPNG 135
AQP4.PRO     TVAMVCTRKISIAKSVFYITAQCLGAIIGAGILYLVTPPSVVGGLG 146
AQP5.PRO     TLALLIGNQISLLRAVFYVAAQLVGAIAGAGILYWLAPLNARGNLA 118
conservation ! * *    ** *****!  !! *!!* ***!*   **    * **
```

For exploratory documents, `motif-map(auto)` can discover common motifs and focus the displayed region automatically.

8.3 Functional Shading

Functional shading groups residues by biochemical properties such as charge, hydropathy, chemical class, or side-chain area.

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    functional("charge"),
    window(1, "138..170"),
    graph("top", 1, "138..170", "charge", kind: "color", options: ("ColdHot",)),
    legend(),
  ),
)
```

Typeset result

```
acidic (-)
basic (+)

AQP1.PRO  GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO  AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO  GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO  GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO  AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
consensus *  **** ! *!********! ** *  !* **
```

Custom functional modes are built from explicit group definitions:

```
#let acidic-basic = (
  functional("custom"),
  clear-functional-groups(),
  functional-group("acidic (-)", "DE", "White", "Red"),
  functional-group("basic (+)", "HKR", "White", "Blue"),
  shade-all-residues(),
)

#let alignment = read("AQPpro.MSF", encoding: none)

#shade(alignment, format: "msf", commands: acidic-basic)
```

Typeset result

```
acidic (-)
basic (+)
```

```
AQP1.PRO   GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO   AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO   GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO   GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO   AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
consensus  ********!**!********!*******!****
```

8.4 Sequence Logo Without Sequence Rows

To emphasize information content rather than individual rows, hide the sequences and keep the logo, conservation row, and ruler.

```
#let alignment = read("AQPpro.MSF", encoding: none)
#shade(
  alignment,
  format: "msf",
  figure: logo-analysis(
    region: "203..235",
    colors: "charge",
    conservation: true,
    commands: (
      hide-all-sequences(),
      ruler("bottom", sequence: 3, every: 1),
    ),
  ),
)
```

Typeset result


```
consensus  *!***!!!!****** *    *         *!
               |    |    |    |    |    |    |
              205  210  215  220  225  230  235
```

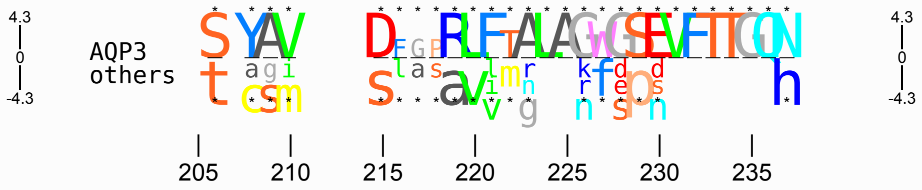
8.5 Subfamily Logo

Subfamily logos compare a selected set of sequences against the remaining sequences. Relevance markers help identify positions with interpretable deviation.

```
#let alignment = read("AQPpro.MSF", encoding: none)

#shade(
  alignment,
  format: "msf",
  figure: logo-analysis(
    sequence: 3,
    region: "203..235",
    subfamily: (3,),
    negative: true,
    relevance: (threshold: 1.0, char: "*", color: "Black"),
    commands: (
      hide-all-sequences(),
      subfamily-logo-name("AQP3", negative-name: "others"),
    ),
  ),
)
```

Typeset result

```
consensus *!***!!!!*******  *     *          *!
```



8.6 Structure And Topology Tracks

Structure tracks can be attached either manually or through `structure-map`.

```
#let alignment = read("AQPpro.MSF", encoding: none)
#let topology = read("AQP1.top", encoding: none)
#let secondary = read("AQP1.phd", encoding: none)

#shade(
  alignment,
  format: "msf",
  figure: structure-map(
    1,
    topology: topology,
    secondary: secondary,
    region: "1..120",
  ),
)
```

Typeset result

```
                   10        20        30        40
                    |         |         |         |
AQP1.PRO    1 MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLE 40
AQP2.PRO    1 MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...S 36
AQP3.PRO    1 M........LLRQALAECLGTLILVMFGCGSVAQVVLSRG 44
AQP4.PRO    1 MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...G 61
AQP5.PRO    1 MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...S 37
consensus     !  *    ** *** !!*!*****!*** !!** *

                   50        60        70        80
                    |         |         |         |
AQP1.PRO   41 RNQTLVQDNVKVSLAFGLSIATLAQSVGHISGAHSNPAVT 80
AQP2.PRO   37 S....PPSVLQIAVAFGLGIGILVQALGHVSGAHINPAVT 72
AQP3.PRO   45 TH....GGFLTINLAFGFAVTLAILVAGQVSGAHLNPAVT 80
AQP4.PRO   62 SENPLPVDMVLISLCFGLSIATMVQCFGHISGGHINPAVT 101
AQP5.PRO   38 A....LPTILQISIAFGLAIGTLAQALGPVSGGHINPAIT 73
consensus              * ****!!* * ** *  !**!!*!*!!!*!

                   90       100       110       120
                    |         |         |         |
AQP1.PRO   81 LGLLLSCQISILRAVMYIIAQCVGAIVASAILSGITSSLL 120
AQP2.PRO   73 VACLVGCHVSFLRAAFYVAAQLLGAVAGAAILHEITPVEI 112
AQP3.PRO   81 FAMCFLAREPWIKLPIYTLAQTLGAFLGAGIVFGLNELVV 130
AQP4.PRO  102 VAMVCTRKISIAKSVFYITAQCLGAIIGAGILYLVTPPSV 141
AQP5.PRO   74 LALLIGNQISLLRAVFYVAAQLVGAIAGAGILYWLAPLNA 113
consensus      * *    ** *****!  !! *!!* ***!*   **
```

When exact filtering is needed, add controls:

```
#let alignment = read("AQPpro.MSF", encoding: none)
#let topology = read("AQP1.top", encoding: none)

#shade(
  alignment,
  format: "msf",
```

```
  commands: (
    structures(1, topology: topology),
    show-structure-types(
      "PHDtopo",
      ("TM", "internal", "external"),
    ),
    structure-appearance(
      "PHDtopo",
      "TM",
      "top",
      "box[Blue]:TM",
      "",
    ),
  ),
)
```

Typeset result

```
AQP1.PRO  MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLERNQTLVQDNVKVSLAFGLSI 60
AQP2.PRO  MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...SS....PPSVLQIAVAFGLGI 52
AQP3.PRO  M........LLRQALAECLGTLILVMFGCGSVAQVVLSRGTH....GGFLTINLAFGFAV 60
AQP4.PRO  MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...GSENPLPVDMVLISLCFGLSI 81
AQP5.PRO  MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...SA....LPTILQISIAFGLAI 53
consensus !  *    ** *** !!*!*****!*** !!** *              * ****!!* *

AQP1.PRO  ATLAQSVGHISGAHSNPAVT 80
AQP2.PRO  GILVQALGHVSGAHINPAVT 72
AQP3.PRO  TLAILVAGQVSGAHLNPAVT 80
AQP4.PRO  ATMVQCFGHISGGHINPAVT 101
AQP5.PRO  GTLAQALGPVSGGHINPAIT 73
consensus  ** *  !**!!*!*!!!*!
```

8.7 PDB-Based Selection

PDB selections are especially useful for showing residues near a structural site without hand-transcribing residue numbers.

```
#let alignment = read("AQPpro.MSF", encoding: none)
#let pdb = read("1J4N.pdb", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    window(1, pdb-point(pdb, 81, distance: 8, atom: "CA")),
    ruler("top", sequence: 1, every: 1),
    highlight(1, pdb-line(pdb, 81, 168), bg: "Yellow"),
  ),
)
```

Typeset result

```
      123
      |||
Alpha AEF. 3
Beta  ADF. 3
Gamma A.G. 2
seq4  AC.. 2
```

8.8 Translation And Complement Features

Single-sequence nucleotide figures can combine lowercased UTRs, complements, translations, and manual feature labels.

```
#let alignment = read("AQPDNA.MSF", encoding: none)
```

```
#shade(
  alignment,
  format: "msf",
  seq-type: "N",
  commands: (
    single-sequence(sequence: 1),
    shift-single-sequence(),
    window(1, "-84..525", start: -84),
    lower(1, "-84..-1,439..525"),
    backtranslation-label("horizontal"),
    mark("bottom", 1, "390..402", fill: "Blue", text: "poly-Arg"),
  ),
)
```

Typeset result

AQP1nuc.SEQ CCTGGGCATTGAGATCATTGGCACCctgca 443

9 Public API Layers

Typshade has three layers. You can mix them, but most documents should start at the top.

Layer	Use When	Examples
`figure:` recipes	You know the purpose of the figure.	`publication`, `motif-map`, `structure-map`, `logo-analysis`
`commands:` helpers	You want reusable pieces or small adjustments.	`similar`, `lines`, `ruler`, `logo`, `highlight`
Fine-grained commands	You need precise control over a TeXshade-equivalent feature.	`weight-table`, `residue-style`, `feature-rule`, `structure-appearance`

10 Figure Recipes

`figure:` is the recommended API for hand-written scientific figures. Recipes expand into lower-level track, annotation, style, and scoring commands.

Recipe	Purpose	Typical Use
`publication`	Balanced paper-ready alignment figure.	`publication(motifs: auto, logo: auto)`
`motif-map`	Detect or highlight motifs and add a conservation graph.	`motif-map(auto)`
`structure-map`	Combine alignment, ruler, consensus, and structure tracks.	`structure-map(1, topology: read("AQP1.top", encoding: none))`
`logo-analysis`	Show sequence/subfamily logo analysis.	`logo-analysis(colors: "charge", subfamily: (1, 2, 3))`
`overview`	Compact overview for many sequences or quick reports.	`overview(colors: "grays")`

Smart defaults use the alignment data:

- `motif-map(auto)` detects common protein or nucleotide motifs.
- `region: auto` focuses around detected motifs.
- `line-length: auto` adapts to sequence count and selected region.
- `logo: auto` enables logos only when they remain readable.
- `threshold: auto` chooses a sequence-type-aware default.

10.1 Example: Motif Map

```
#let alignment = read("alignment.msf", encoding: none)

#shade(
  alignment,
  format: "msf",
```

```
  figure: motif-map(
    (
      "NPA": (bg: "BrickRed", text: "active site"),
      "NXX[ST]N": "glycosylation",
    ),
    region: "80..125",
    logo: "charge",
  ),
)
```

Typeset result


```
              80                  90                  100                 110                 120
              |                   |                   |                   |                   |
AQP1.PRO      TLGLLLSCQISILRAVMYIIAQCVGAIVASAILSGITSSLLENSLG 125
AQP2.PRO      TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAILHEITPVEIRGDLA 117
AQP3.PRO      TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIVFGLNELVVSGPNG 135
AQP4.PRO      TVAMVCTRKISIAKSVFYITAQCLGAIIGAGILYLVTPPSVVGGLG 146
AQP5.PRO      TLALLIGNQISLLRAVFYVAAQLVGAIAGAGILYWLAPLNARGNLA 118
consensus     ! * *    ** *****!  !! *!!* ***!*   **    * **
```

11 Command Helpers

commands: is useful when a recipe is too high-level and you want to assemble the visible parts yourself. It accepts helper output and low-level commands in left-to-right order.

Helper	Purpose	Example
similar, identical, diverse, functional	Choose scoring and optional colors/thresholds.	similar(colors: "blues", threshold: 45)
lines	Set residues per line.	lines(50)
window	Show a selected sequence range.	window(1, "80..125")
names, numbers	Configure sequence labels and numbering.	names(color: "RoyalBlue")
ruler, consensus, logo, legend	Add common tracks.	ruler("top", every: 10)
highlight, tint, emphasize	Style residue regions.	highlight(1, "NPA", bg: "BrickRed")
motif, mark, graph	Add feature rows and graph tracks.	motif(1, "NXX[ST]N", text: "motif")
structures	Add topology/secondary-structure tracks.	structures(1, topology: read("AQP1.top", encoding: none))
typography	Adjust text for named parts.	typography(target: "names", family: "mono")

11.1 Example: Publication Figure

```
#let alignment = read("alignment.msf", encoding: none)

#shade(
  alignment,
  format: "msf",
  preset: "publication",
  theme: "screen",
  commands: (
    similar(threshold: 45),
    lines(50),
    ruler("top", sequence: 1, every: 10),
    consensus("bottom", name: "conservation"),
```

```
  ),
)
```

Typeset result

```
               80        90        100       110       120
               |         |         |         |         |
AQP1.PRO       TLGLLLSCQISILRAVMYIIAQCVGAIVASAILSGITSSLLENSLG 125
AQP2.PRO       TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAILHEITPVEIRGDLA 117
AQP3.PRO       TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIVFGLNELVVSGPNG 135
AQP4.PRO       TVAMVCTRKISIAKSVFYITAQCLGAIIGAGILYLVTPPSVVGGLG 146
AQP5.PRO       TLALLIGNQISLLRAVFYVAAQLVGAIAGAGILYWLAPLNARGNLA 118
conservation   ! * *    ** *****!  !! *!!* ***!*   **    * **
```

11.2 Example: Motifs And Graphs

```
#let alignment = read("alignment.msf", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    window(1, "80..125"),
    highlight(1, "NPA", bg: "BrickRed"),
    motif(1, "NXX[ST]N", text: "glycosylation"),
    graph("bottom", 1, "all", "conservation", kind: "color", options: ("ColdHot",)),
  ),
)
```

Typeset result

```
AQP1.PRO   TLGLLLSCQISILRAVMYIIAQCVGAIVASAILSGITSSLLENSLG 125
AQP2.PRO   TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAILHEITPVEIRGDLA 117
AQP3.PRO   TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIVFGLNELVVSGPNG 135
AQP4.PRO   TVAMVCTRKISIAKSVFYITAQCLGAIIGAGILYLVTPPSVVGGLG 146
AQP5.PRO   TLALLIGNQISLLRAVFYVAAQLVGAIAGAGILYWLAPLNARGNLA 118
consensus  ! * *    ** *****!  !! *!!* ***!*   **    * **
```

12 Helper Function API

Helper functions are useful when a figure is assembled from reusable fragments.

```
#let active-site = (
  highlight(1, "NPA", fg: "White", bg: "BrickRed"),
  motif(1, "NXX[ST]N", text: "motif"),
)

#shade(
  read("alignment.msf", encoding: none),
  format: "msf",
  preset: "publication",
  theme: "screen",
  regions: active-site,
  commands: (
    scoring-mode("similar"),
    ruler-track(position: "top", sequence: 1, steps: 10),
  ),
)
```

Typeset result

```
          80        90        100       110       120
          |         |         |         |         |
AQP1.PRO  TLGLLLSCQISILRAVMYIIAQCVGAIVASAILSGITSSLLENSLG 125
AQP2.PRO  TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAILHEITPVEIRGDLA 117
AQP3.PRO  TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIVFGLNELVVSGPNG 135
AQP4.PRO  TVAMVCTRKISIAKSVFYITAQCLGAIIGAGILYLVTPPSVVGGLG 146
AQP5.PRO  TLALLIGNQISLLRAVFYVAAQLVGAIAGAGILYWLAPLNARGNLA 118
consensus ! * *    ** *****!  !! *!!* ***!*   **    * **
```

13 Detailed User Guide

This section describes the public surface in the style of a user manual. The tables below are intentionally redundant with the compatibility reference: the guide explains how to write Typshade; the compatibility reference explains where each TeXshade command went.

13.1 The `shade` Function

`shade` is the only rendering function most documents need.

The following block is a signature reference, not runnable sample code:

```
#shade(
  source,
  format: auto,
  figure: (),
  preset: none,
  theme: none,
  mode: none,
  option: none,
  seq-type: auto,
  residues-per-line: none,
  fit: none,
  names: none,
  numbering: none,
  consensus: none,
  ruler: none,
  logo: none,
  subfamily-logo: none,
  legend: none,
  regions: (),
  features: (),
  commands: (),
  caption: none,
  short-caption: none,
  font: none,
  font-size: none,
)
```

Argument	Meaning
`source`	Alignment source, normally `read("alignment.msf", encoding: none)`. This is the only required argument.
`format`	`auto` by default; use explicit formats for ambiguous MSF/ALN/FASTA-like files.
`figure`	Purpose-level recipe or array of recipes/commands. This is the recommended high-level API.
`preset`	Named reusable defaults such as publication/overview-oriented settings.
`theme`	Named or custom visual theme. Themes should express visual identity, not biology.
`mode`, `option`	Direct scoring-mode override. Prefer `similar`, `functional`, `tcoffee`, etc. inside `commands:` when assembling manually.
`seq-type`	`auto`, `"P"`, or `"N"` depending on whether type should be inferred or forced.
`residues-per-line`, `fit`	Direct line length override, or Typst-aware fitting. Use `fit: "container"` for width-based layout and `fit: "page"` to also enable page-aware block splitting. Shortcut equivalent: `lines(count)` or `lines(auto)`.
`names`, `numbering`, `consensus`, `ruler`, `logo`, `legend`	Convenience toggles or simple track declarations. Use track helpers for more control.

`regions`, `features`	Dedicated slots for highlight/annotation command fragments.
`commands`	Final explicit command list. Use this for low-level controls and final overrides.
`caption`, `short-caption`	`caption` wraps the rendered alignment in a Typst `figure`; `short-caption` stores companion metadata for workflows that need it.
`font`, `font-size`	Renderer-level fallback text settings. Prefer `typography` for targeted styling.

By default, rendered alignment blocks are left-aligned. Use `alignment-position("center")` or `alignment-position("right")` only when a particular figure should be optically centered or flush-right inside its containing block.

13.2 Recipe Options

Recipes are the main UI/UX improvement over TeXshade. They describe the kind of figure rather than the macro operations needed to build it.

`publication` option	Meaning
`mode`	Scoring mode, default `"similar"`.
`similarity`	Color scheme, default `"blues"`.
`threshold`	`auto` chooses a sequence-type-aware threshold; a number fixes it.
`sequence`	Reference sequence for region, ruler, and motif interpretation.
`region`	`auto`, `none`, range string, motif string, or dictionary with `sequence`/`selection`.
`line-length`	`auto` picks a readable line length; a number fixes residues per line.
`ruler`, `every`	Enable ruler and choose tick interval.
`conservation`	Controls consensus/conservation row.
`logo`	Logo color set, `auto`, `false`, or `none`.
`motifs`, `highlights`	Motif and region annotations.
`theme`, `annotations`, `commands`	Additional visual theme and final command fragments.

Recipe	Distinctive Options
`motif-map`	`motifs`, `region`, `graph`, `logo`, and `conservation` can all be `auto` so the alignment drives the result.
`structure-map`	`topology`, `secondary`, `hmmtop`, and `hmmtop-sequence` attach structure tracks to a chosen sequence.
`logo-analysis`	`colors`, `subfamily`, `negative`, and `relevance` configure sequence/subfamily logo analysis.
`overview`	`names`, `numbers`, `conservation`, and `ruler` default to compact values for large alignments.

13.3 Selection Syntax

Selections are used by windows, highlights, motifs, feature rows, graphs, and PDB helpers.

Selection Kind	Example	Meaning
Single range	`"80..112"`	Residues 80 through 112 in the reference sequence.
Multiple ranges	`"80..90,100..110"`	Union of several displayed spans.
Keyword	`"all"`	All available positions in the selected sequence.
Motif	`"NPA"`	All matching motif occurrences.
Pattern motif	`"NXX[ST]N"`	`X` matches any residue; bracket groups match any listed residue.
Selection DSL	`select-or(select-motif("NPA"), select-metric("coverage", at-least: 80))`	Composable selection objects for ranges, motifs, metrics, negation, intersections, unions, and padding.
PDB point	`pdb-point(read("1J4N.pdb", encoding: none), 81, distance: 8, atom: "CA")`	Residues near a point around a PDB anchor.
PDB line	`pdb-line(read("1J4N.pdb", encoding: none), 81, 168)`	Residues close to a line between two anchors.
PDB plane	`pdb-plane(read("1J4N.pdb", encoding: none), 66, 73, 199)`	Residues close to a plane defined by three anchors.

Use `selection-preview` or `selection-table` when a motif, metric, PDB, or composed DSL selection is too subtle to trust by eye.

13.4 Feature Rows

The general TeXshade `\feature` command is powerful but dense. Typshade splits common cases into clearer helpers:

Task	Helper	Example
Text or boxed mark	`mark`	`mark("top", 1, "93..93", fill: "Yellow", text: "site")`
Motif highlight plus label	`motif`	`motif(1, "NXX[ST]N", text: "glycosylation")`
Region color	`highlight`	`highlight(1, "NPA", fg: "White", bg: "BrickRed")`
Tint existing shading	`tint`	`tint(1, "158..163", intensity: "strong")`
Emphasize text	`emphasize`	`emphasize(1, "QLVLC", style: "italic")`
Graph track	`graph`	`graph("bottom", 1, "all", "conservation", kind: "color")`

Built-in graph metrics include `conservation`, `entropy`, `gap-fraction`, `coverage`, `identity`, `hydrophobicity`, `molweight`, and `charge`. Use external graph data only when the metric is not already derivable from the alignment itself.

When you need exact TeXshade-style label strings, use the lower-level feature controls through `commands:`. The renderer still supports feature positions such as `ttttop`, `tttop`, `ttop`, `top`, `bottom`, `bbottom`, `bbbottom`, and `bbbbottom`.

13.5 Colors And Palettes

Typshade accepts the package color names used by its palettes and resolves them through Typst colors. Built-in color schemes include TeXshade-style schemes such as `blues`, `reds`, `greens`, `grays`, and `black`, plus graph scales such as `BlueRed`, `RedBlue`, `GreenRed`, `RedGreen`, `ColdHot`, `HotCold`, `WhiteBlack`, and `BlackWhite`.

Use these levels of control:

1. `theme:` for the document's visual direction.
2. `color-scheme` or scoring shortcuts for alignment categories.
3. `residue-style`, `cell-style`, `functional-style`, `logo-color`, or `consensus-colors` for exact category/residue styling.
4. `visual-theme` when creating a reusable house style.

`cell-style` accepts a Typst function and is evaluated for each rendered sequence cell after built-in scoring and before explicit region annotations:

```
#shade(
  alignment,
  format: "msf",
  commands: (
    cell-style(ctx => if ctx.at("consensus-score") < 70 {
      (frame: "BrickRed")
    } else {
      none
    }),
  ),
)
```

Typeset result Cell-style callback output

```
AQP1.PRO TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
```

13.6 Typography And Layout

TeXshade exposes many short font commands because TeX font selection is macro oriented. Typshade uses targeted text controls:

Control	Purpose
`typography(target: ..., family: ..., weight: ..., posture: ..., size: ...)`	Convenient combined text styling.
`text-family`, `text-weight`, `text-posture`, `text-size`	Set one text attribute for one target.
`text-style`	Set family, weight, posture, and size together.
`character-stretch`, `line-stretch`	Tune residue cell dimensions.
`auto-layout`, `auto-page`, `fit:`	Let Typst choose line lengths from the current container and optionally insert page-aware block breaks.
`block-gap`, `line-gap`, `feature-slot-space`	Tune vertical spacing.
`caption`, `short-caption`	`caption` attaches visible figure-style captions; `short-caption` stores compatibility metadata for external workflows.

13.7 Pairwise Similarity And Identity

TeXshade exposes `\percentsimilarity`, `\percentidentity`, and `\similaritytable` outside the alignment environment. Typshade provides the same analysis as explicit data/document helpers:

```
#let alignment = read("AQPpro.MSF", encoding: none)

#percent-similarity(alignment, 1, 2, format: "msf", selection: "80..112")
#percent-identity(alignment, "AQP1.PRO", "AQP2.PRO", format: "msf")
#similarity-table(alignment, format: "msf", selection: "80..112")
```

Typeset result Pairwise identity and similarity

Metric	Value
`percent-similarity(1, 2)`	72.7 + “%”
`percent-identity(AQP1, AQP2)`	44.9 + “%”

	AQP1.PRO	AQP2.PRO	AQP3.PRO	AQP4.PRO	AQP5.PRO
AQP1.PRO	-	72.7	57.6	81.8	81.8
AQP2.PRO	42.4	-	54.5	69.7	75.8
AQP3.PRO	21.2	33.3	-	51.5	48.5
AQP4.PRO	57.6	51.5	33.3	-	75.8
AQP5.PRO	48.5	57.6	33.3	51.5	-

`similarity-table` follows TeXshade’s combined table convention: values above the diagonal are percent similarity and values below the diagonal are percent identity. Calculations use shared non-gap positions in the selected alignment span.

14 TeXshade Compatibility Reference

This section is intentionally detailed. It is the migration contract: every major TeXshade feature is listed with the corresponding Typshade API and the reason when the mapping is not literal.

Typshade does not keep TeX macro names as public names. Instead it preserves the scientific capability and exposes it through Typst-style names, typed values, recipes, and explicit command lists.

Status	Meaning

`native`	The feature is available through a Typst-native API and should be preferred for new documents.
`control`	The feature is available as a lower-level command helper, usually inside `commands:`.
`recipe`	The feature is normally expressed through a high-level `figure:` recipe.
`data`	The TeXshade command produced messages or files; Typshade exposes data/content instead.
`approx`	The visible behavior is implemented, but TeX box/glue details are intentionally not reproduced exactly.
`omitted`	The behavior is a TeX side effect, external file writer, or driver-specific hook that is not appropriate inside a Typst package.

14.1 Reading The Migration Tables

Each table row gives the TeXshade command or feature, the Typshade replacement, the support status, and migration notes. When a row lists several TeXshade commands, those commands are one feature family in TeXshade and are configured together in Typshade.

For new documents, prefer this order:

1. Use `figure:` recipes when the figure has a recognizable purpose.
2. Use shortcut helpers such as `similar`, `ruler`, `logo`, `motif`, and `structures` when you are assembling a figure manually.
3. Use lower-level controls such as `weight-table`, `residue-style`, `structure-appearance`, or `feature-slot-space` only when you need exact control over a TeXshade-equivalent detail.

14.2 Core Environment And Input

TeXshade	Typshade	Status	Migration Notes
`\begin{texshade}{file}...\end{texshade}`	`shade(read("file", encoding: none), format: "msf", figure: publication(...))`	`native`	The entry point is a function call. The alignment source is explicit Typst data supplied by the caller, so package rendering does not depend on package-internal file access.
TeXshade parameter files	`#let common = (...)` plus `commands: common` or `figure: ...`	`native`	Use Typst variables, dictionaries, arrays, imports, and presets instead of order-sensitive parameter files.
`seqtype`	`seq-type:` or `sequence-type("P")` / `sequence-type("N")`	`native`	Automatic detection remains available; explicit overrides are stable and local to the figure.
MSF, FASTA-like, ALN-like inputs	`format: auto`, `format: "msf"`, or explicit parser format	`native`	Use `format: auto` unless an ambiguous file needs an override.
TeX command stream inside the environment	`commands: (...)`, `regions: (...)`, `features: (...)`, `figure: ...`	`native`	The order is visible as Typst array order; reusable command fragments can be normal variables.
TeXshade inspection by compiling/log output	`alignment-summary`, `sequence-list`, `selection-preview`, `selection-table`	`data`	Inspection is document content/data and can be embedded in reports.

14.3 Recommended Figure-Level Replacements

TeXshade Use Case	Typshade	Status	Migration Notes
Publication alignment with shading, ruler, consensus	`figure: publication(...)`	`recipe`	This replaces a long TeXshade setup with a purpose-level declaration. Use `similarity`, `threshold`, `region`, `motifs`, `logo`, and `ruler` options.
Motif-centric figure	`figure: motif-map(auto)` or `motif-map((...))`	`recipe`	Automatically detects common motifs, can focus the displayed region, and can add motif labels and conservation graphs.
Structure/topology figure	`figure: structure-map(sequence, topology: ..., secondary: ...)`	`recipe`	Combines alignment, ruler, consensus, and topology/secondary-structure tracks.
Logo/subfamily analysis	`figure: logo-analysis(subfamily: (...), colors: ...)`	`recipe`	Builds sequence-logo and subfamily-logo views without manually managing all logo commands.
Dense overview of many sequences	`figure: overview(...)`	`recipe`	Compact defaults for high sequence counts or report appendices.

14.4 Shading, Similarity, And Scoring

TeXshade	Typshade	Status	Migration Notes
`shadingmode[similar]{identical}`	`identical(...)`, `similar(...)`, `diverse(...)`, `functional(...)`, `single-sequence(...)`, or `scoring-mode(...)`	`native`	The common modes are shortcuts; `scoring-mode` is the exact lower-level control. T-Coffee uses `tcoffee(...)` or `tcoffee-scores(...)`.
`shadingcolors`	`color-scheme("blues")` or `similar(colors: "blues")`	`native`	Color scheme selection is separated from scoring mode so themes can be reused.
`defshadingcolors`	`shade-theme(...)`, `visual-theme(...)`, `color-scheme(...)`, and `residue-style(...)`	`native`	Rather than snapshotting mutable TeX state, Typshade encourages named dictionaries and explicit style commands.
`nomatchresidues`, `similarresidues`, `conservedresidues`, `allmatchresidues`	`residue-style("nomatch", ...)`, `residue-style("similar", ...)`, `residue-style("conserved", ...)`, `residue-style("allmatch", ...)`	`control`	Foreground, background, case, and emphasis style can be configured per residue category.
`threshold`	`threshold(45)` or `similar(threshold: 45)`	`native`	Controls conservation and shading threshold. Recipes accept `threshold: auto` for sequence-type-aware defaults.
`allmatchspecial`, `allmatchspecialoff`	`all-match-threshold(value: 100)`, `disable-all-match-threshold()`	`control`	Controls whether very high-conservation columns receive special handling.
`hideallmatchpositions`, `showallmatchpositions`	`hide-all-match-positions()`, `show-all-match-positions()`	`control`	Supports removal/restoration of columns above the all-match threshold.
`shadeallresidues`	`shade-all-residues()`	`control`	Equivalent to lowering the threshold so every residue is eligible for shading.
`weighttable`	`weight-table("BLOSUM62")`, `weight-table("PAM250")`, `weight-table("PAM100")`, `weight-table("structural")`	`control`	Weighted consensus scoring is available with common TeXshade matrices.
`setweight`	`set-weight("A", "G", value)`	`control`	Overrides individual pair weights for custom scoring.
`gappenalty`	`gap-penalty(value)`	`control`	Controls how gaps contribute to weighted consensus scoring.
`pepgroups`, `pepsims`	`peptide-groups(...)`, `peptide-similarities(...)`	`control`	Custom peptide groupings and per-residue similarity sets.
`DNAgroups`, `DNAsims`	`dna-groups(...)`, `dna-similarities(...)`	`control`	Custom nucleotide groupings and per-base similarity sets.
`clearfuncgroups`, `funcgroup`	`clear-functional-groups()`, `functional-group(...)`	`control`	Defines a custom functional shading scheme.
`funcshadingstyle`	`functional-style(residue, fg, bg, case: ..., style: ...)`	`control`	Overrides style for a functional residue category.
`T-Coffee` scoring file option, `includeTCoffee`	`tcoffee(read("scores.tcs", encoding: none))` or `tcoffee-scores(read("scores.tcs", encoding: none))`	`native`	Loads T-Coffee confidence data as a scoring mode option.

14.5 Sequence Lines, Labels, Numbering, Gaps

TeXshade	Typshade	Status	Migration Notes
`residuesperline`, `residuesperline*`	`lines(60)`, `lines(auto)`, `residues-per-line(auto)`, `fit: "container"`, or `auto-layout(...)`	`native`	Controls columns per alignment block. Auto layout uses Typst layout information to fit the current container.
Manual page-break tuning	`fit: "page"` or `auto-page(blocks: auto)`	`native`	Typst can split long alignment blocks using page-aware layout and optional explicit block counts.
`setends`	`window(sequence, "80..125", start: ...)`, `sequence-window(...)`, or Selection DSL values	`native`	Selects displayed ranges by sequence and residue selection. DSL values can combine ranges, motifs, metrics, negation, and padding.
`setdomain`	`domain(sequence, selection)`	`control`	Domain selection maps to the same explicit window model. Domain-specific gap styling uses normal gap controls.

`shownames`, `hidenames`	`names(position: "left")`, `no-names()`	`native`	Sequence-name labels are an explicit track.
`nameseq`	`sequence-name(sequence, "label")`	`control`	Renames an individual sequence label.
`namescolor`, `namecolor`	`names-color(color)`, `sequence-name-color(sequences, color)`	`control`	Global and per-sequence label colors.
`hidename`	`hide-sequence-name(sequences)`	`control`	Hides selected sequence labels while keeping rows.
`shownumbering`, `hidenumbering`	`numbers(position: "right")`, `no-numbers()`	`native`	Numbering is an explicit track.
`numberingcolor`, `numbercolor`	`numbering-color(color)`, `sequence-number-color(sequences, color)`	`control`	Global and per-sequence numbering colors.
`hidenumber`	`hide-sequence-number(sequences)`	`control`	Hides numbering for selected rows.
`startnumber`	`start-number(sequence, start, selection: ...)`	`control`	Overrides displayed numbering start and optional selection scope.
`allowzero`, `disallowzero`	`allow-zero-numbering()`, `disallow-zero-numbering()`	`control`	Controls whether zero can appear in numbering.
`seqlength`	`sequence-length(sequence, length)`	`control`	Overrides reported sequence length.
`gapchar`	`gap-char(symbol)`	`control`	Changes visible gap symbol.
`gaprule`	`gap-rule(thickness)`	`control`	Draws gaps as rules.
`gapcolors`	`gap-colors(foreground, background)`	`control`	Sets rule/character gap colors.
`domaingaprule`, `domaingapcolors`	`domain-gap-rule(thickness)`, `domain-gap-colors(foreground, background)`	`control`	Available as domain-oriented aliases over the same gap styling model.
`showleadinggaps`, `hideleadinggaps`	`show-leading-gaps()`, `hide-leading-gaps()`	`control`	Controls whether leading gaps are visible.
`stopchar`	`stop-char(symbol)`	`control`	Custom stop codon/terminal symbol.
`shiftsingleseq`, `keepsingleseqgaps`	`shift-single-sequence(value: ...)`, `keep-single-sequence-gaps()`	`approx`	Single-sequence offset behavior is represented, but TeX's unusual gap-rewriting edge cases are not treated as byte-for-byte compatibility targets.
`hideresidues`, `showresidues`	`hide-residues()`, `show-residues()`	`control`	Hides residue letters while preserving colored cells.
`fingerprint`	`fingerprint(value)`	`approx`	Provides condensed fingerprint-like rendering controls; exact TeX ultra-thin bar metrics are Typst layout approximations.

14.6 Consensus, Rulers, Logos, And Legend

TeXshade	Typshade	Status	Migration Notes
`showconsensus`, `hideconsensus`	`consensus(position: "bottom")`, `consensus-track(...)`, `no-consensus()`	`native`	Adds or removes the consensus row.
`nameconsensus`	`consensus-name("consensus")`	`control`	Controls consensus row label.
`defconsensus`	`consensus-symbols(none, conserved, allmatch)`	`control`	Controls symbols used in the consensus row.
`consensuscolors`	`consensus-colors(...)`	`control`	Configures foreground/background colors for consensus states.
`constosingleseq`, `constoallseqs`	`consensus-from-sequence(sequence)`, `consensus-from-all-sequences()`	`control`	Controls the basis for consensus calculation.
`germanlanguage`, `spanishlanguage`, `englishlanguage`	`consensus-language("german")`, etc.	`control`	Applies to document-facing consensus labels. TeX diagnostic message localization is not reproduced.
`exportconsensus`	`alignment-data(...)`, `parse-alignment(...)`, or external tooling	`omitted`	Typst packages should not write arbitrary external files while rendering. Export-like workflows should use explicit build scripts.
`showruler`, `hideruler`	`ruler("top", sequence: 1)`, `ruler-track(...)`, `no-ruler(...)`	`native`	Adds top/bottom ruler tracks.

`rulersteps`	`ruler-steps(value, position: ...)` or `ruler(..., every: value)`	`native`	Controls tick interval.
`rulercolor`	`ruler-color(color, position: ...)`	`control`	Controls ruler color.
`namerulerpos`, `nameruler`	`ruler-marker(number, text, position: ...)`, `ruler-name(name, position: ...)`	`control`	Adds named ruler markers or a ruler label.
`rulernamecolor`, `rulerspace`	`ruler-name-color(color)`, `ruler-space(value)`	`control`	Controls ruler label styling and spacing.
`rotateruler`, `unrotateruler`	`rotate-ruler(position: ...)`, `unrotate-ruler(position: ...)`	`approx`	Typst rotates content, but TeX baseline/glue side effects are not copied exactly.
`showsequencelogo`, `hidesequencelogo`	`logo(position: "top")`, `sequence-logo(...)`, `no-sequence-logo()`	`native`	Adds or removes sequence logos.
`namesequencelogo`	`sequence-logo-name(name)` or `sequence-logo(name: ...)`	`control`	Controls sequence-logo label.
`showlogoscale`, `hidelogoscale`	`logo-scale(position: ..., color: ...)`, `no-logo-scale()`	`control`	Controls logo scale visibility and placement.
`logostretch`	`logo-stretch(value)`	`control`	Vertical stretch for logo display.
`logocolor`, `clearlogocolors`	`logo-color(residues, color)`, `clear-logo-colors(default: ...)`	`control`	Per-residue logo color control.
`dofrequencycorrection`, `undofrequencycorrection`	`frequency-correction()`, `no-frequency-correction()`	`control`	Controls small-sample correction for logo information content.
`setsubfamily`, `showsubfamilylogo`, `hidesubfamilylogo`	`subfamily(sequences)`, `subfamily-logo(...)`, `no-subfamily-logo()`	`native`	Subfamily logo support is available either as explicit commands or through `logo-analysis`.
`namesubfamilylogo`	`subfamily-logo-name(name, negative-name: ...)`	`control`	Controls labels for positive/negative subfamily logos.
`shownegatives`, `hidenegatives`	`negative-logo-values()`, `no-negative-logo-values()`	`control`	Controls negative logo values.
`relevance`, `subfamilythreshold`, `showrelevance`, `hiderelevance`	`relevance-threshold(value)`, `relevance-marker(char: ..., color: ...)`, `no-relevance-marker()`	`control`	Controls subfamily-logo relevance marks. `subfamilythreshold` is treated as an older spelling for the same threshold concept.
`showlegend`, `hidelegend`	`legend(color: ...)`, `legend-track(...)`, `no-legend()`	`native`	Legend entries derive from functional groups or the actually rendered residue style categories.
`legendcolor`, `movelegend`, `shadebox`	`legend-color(color)`, `legend-offset(dx, dy)`, `color-swatch(color)`	`control`	Legend placement and swatches are Typst content rather than TeX boxes.

14.7 Regions, Motifs, Features, Graphs, Translation

TeXshade	Typshade	Status	Migration Notes
`shaderegion`, `shadeblock`	`highlight(sequence, selection, ...)` or `highlight-block(...)`	`native`	Selections support ranges, motif-like patterns, and PDB-derived selections.
`changeshadingcolors`	`region-color-scheme(sequence, selection, scheme)`	`control`	Changes the color scheme for a selected region.
`tintregion`, `tintblock`	`tint(sequence, selection, intensity: ...)`, `tint-block(...)`	`native`	Applies tinting to selected residues.
`tintdefault`	`tint-default(effect)`	`control`	Controls default tint behavior.
`emphregion`, `emphblock`	`emphasize(sequence, selection, style: ...)`, `emphasis-block(...)`	`native`	Applies emphasis to selected residues.
`emphdefault`	`emphasis-default(style)`	`control`	Controls default emphasis behavior.
`lowerregion`, `lowerblock`	`lower(sequence, selection)`, `lower-block(...)`	`control`	Renders selected residues in lowercase.
`frameblock`	`frame(sequence, selection, color: ...)`	`control`	Frames selected regions.
`feature`	`mark(position, sequence, selection, ...)`, `motif(...)`, `graph(...)`	`native`	Most hand-written features are easier through `mark`, `motif`, and `graph`; raw feature

			styles remain available via lower-level controls.
Feature slots such as `top`, `ttop`, `bottom`, `bbottom`	`mark(position, ...)`, `motif(position: ...)`, `feature-slot-space(position, value)`	`native`	Slot names are preserved as position strings where meaningful.
`ttttopspace`, `tttopspace`, `ttopspace`, `topspace`, `bottomspace`, `bbottomspace`, `bbbottomspace`, `bbbbottomspace`	`feature-slot-space(position, value)`	`native`	The separate TeX spacing macros are replaced by one explicit slot-space control keyed by position.
`featurerule`	`feature-rule(thickness)`	`control`	Controls boxed feature and structure row stroke thickness.
`showfeaturename`, `showfeaturestylename`	`feature-text-label(position, name)`, `feature-style-label(position, name)`	`control`	Names descriptive-text rows and feature-style rows.
`hidefeaturename`, `hidefeaturestylename`, `hidefeaturenames`, `hidefeaturestylenames`	`hide-feature-text-label`, `hide-feature-style-label`, `hide-feature-text-labels`, `hide-feature-style-labels`	`control`	Controls feature label visibility.
`featurenamecolor`, `featurestylenamecolor`, `featurenamescolor`, `featurestylenamescolor`	`feature-text-label-color-at`, `feature-style-label-color-at`, `feature-text-label-color`, `feature-style-label-color`	`control`	Controls feature label colors globally or per slot.
Bar graph feature styles	`graph(position, sequence, selection, metric, kind: "bar", ...)`	`native`	Built-in metrics include conservation, entropy, gap-fraction, coverage, identity, hydrophobicity, molecular weight, and charge.
Color scale feature styles	`graph(..., kind: "color", options: (...))`	`native`	Color-scale tracks support named color ramps and explicit ranges.
Stacked graph styles	`graph(..., kind: "stacked", ...)`	`native`	Stacked rows are available for multi-series feature data.
`bargraphstretch`, `colorscalestretch`	`bar-graph-stretch(value)`, `color-scale-stretch(value)`	`control`	Controls graph track height.
`codon`	`codon(residue, triplets)`	`control`	Custom codon mapping.
`geneticcode`	`genetic-code(name)`	`control`	Selects genetic-code behavior for translation features.
`backtranslabel`, `backtranstext`	`backtranslation-label(...)`, `backtranslation-text(...)`	`approx`	The biological annotation is preserved. Exact TeX box geometry for horizontal/vertical/oblique/zigzag labels is approximated in Typst layout.

14.8 Sequence Visibility, Ordering, Separators, Layout

TeXshade	Typshade	Status	Migration Notes
`hideseq`	`hide-sequence(sequence)`	`control`	Hides one sequence row.
`hideseqs`	`hide-all-sequences()`	`control`	Hides all sequence rows until selectively reintroduced by other settings.
`showseqs`	`show-all-sequences()`	`control`	Restores sequence row visibility.
`killseq`	`remove-sequence(sequence)`	`control`	Removes a sequence from the displayed alignment model.
`donotshade`	`no-shade(sequences)`	`control`	Displays selected sequences without shading.
`orderseqs`	`sequence-order(order)`	`control`	Reorders sequence rows.
`separationline`	`separation-line(sequence)`	`control`	Adds separator after a sequence.
`smallsep`, `medsep`, `bigsep`, `smallsepline`, `medsepline`, `bigsepline`, `nosepline`	`small-separator()`, `medium-separator()`, `large-separator()`, `no-line-gap()`	`control`	Controls separator thickness/spacing. The older `*sepline` aliases are covered by the same explicit spacing helpers.
`numberingwidth`	`numbering-width(digits)`	`control`	Reserves numbering column width.
`charstretch`	`character-stretch(value)`	`control`	Controls cell width/stretch.
`linestretch`	`line-stretch(value)`	`control`	Controls row height/stretch.
`alignment`	`alignment-position(position)`	`control`	Controls alignment placement in its containing block. Typshade defaults to `left`; accepted values are `"left"`, `"center"`, and `"right"`.

`smallblockskip`, `medblockskip`, `bigblockskip`, `noblockskip`	`small-block-gap()`, `medium-block-gap()`, `large-block-gap()`, `no-block-gap()`	`control`	Block gap sizes are deterministic Typst spacing commands.
`vblockspace`	`block-gap(value)`	`control`	Explicit block gap.
`flexblockspace`, `fixblockspace`	`flexible-block-gap()`, `fixed-block-gap()`	`native`	`flexible-block-gap()` inserts weak Typst vertical space between rendered blocks so spacing can disappear cleanly at page boundaries. `fixed-block-gap()` uses fixed `stack` spacing for deterministic block separation.
`vsepspace` and line-gap-like spacing	`line-gap(value)`, `small-line-gap()`, `medium-line-gap()`, `large-line-gap()`, `no-line-gap()`	`control`	Explicit row/line spacing controls.
`alignrightlabels`, `unalignrightlabels`	`align-right-labels()`, `align-left-labels()`	`control`	Controls sequence-label column alignment.
`showcaption`, `shortcaption`	`caption(text, position: ...)`, `short-caption(text)`	`native`	`caption` content is ordinary Typst content and integrates with document layout. `short-caption` is metadata-only compatibility state.

14.9 Structure Tracks And PDB Selections

TeXshade	Typshade	Status	Migration Notes
`includeDSSP`	`dssp-track(sequence, source)`	`native`	Adds DSSP secondary-structure annotation.
`includeSTRIDE`	`stride-track(sequence, source)`	`native`	Adds STRIDE annotation.
`includeHMMTOP`	`hmmtop-track(sequence, source, source-sequence: ...)`	`native`	Adds HMMTOP topology annotation.
`includePHDtopo`	`phd-topology-track(sequence, source)`	`native`	Adds PHD topology annotation.
`includePHDsec`	`phd-secondary-track(sequence, source)`	`native`	Adds PHD secondary-structure annotation.
TeXshade `make new` structure-file modes	External preprocessing, then `dssp-track` / `stride-track` / `hmmtop-track`	`omitted`	Creating new external sources is intentionally outside package rendering. Use a build step, then pass the generated source to Typshade.
`showonDSSP`, `showonSTRIDE`, `showonHMMTOP`, `showonPHDtopo`, `showonPHDsec`	`show-structure-types(format, types)`	`control`	Controls which structure types are visible for a format.
`hideonDSSP`, `hideonSTRIDE`, `hideonHMMTOP`, `hideonPHDtopo`, `hideonPHDsec`	`hide-structure-types(format, types)`	`control`	Hides selected structure types.
`appearance`	`structure-appearance(format, structure-type, position, style, text)`	`control`	Controls style and label text for structure annotations.
`numcount`, `alphacount`, `Alphacount`, `romancount`, `Romancount`	Template placeholders inside `structure-appearance` style/text strings.	`native`	Structure appearance templates expand numeric, alphabetic, and roman counters when rendering repeated structure types.
`firstcolumnDSSP`, `secondcolumnDSSP`	`use-first-dssp-column()`, `use-second-dssp-column()`	`control`	Selects which DSSP structure column is used.
`printPDBlist`, `messagePDBlist`	`pdb-selection(selection)`	`data`	Returns selection-derived residue data/ content instead of writing console messages.
PDB `point`, `line`, `plane` selections	`pdb-point(...)`, `pdb-line(...)`, `pdb-plane(...)`	`native`	Helper functions build readable selection strings for distance-based PDB selections.

14.10 Typography, Themes, And Presets

TeXshade	Typshade	Status	Migration Notes
`setfamily`	`text-family(target, family)` or `typography(family: ...)`	`native`	Targets include alignment parts such as names, numbers, residues, consensus, ruler, and features.

`setseries`	`text-weight(target, weight)` or `typography(weight: ...)`	`native`	Uses Typst's text weight model instead of TeX font series names.
`setshape`	`text-posture(target, posture)` or `typography(posture: ...)`	`native`	Uses Typst posture/style semantics.
`setsize`	`text-size(target, size)` or `typography(size: ...)`	`native`	Can set absolute Typst lengths or package size keywords where supported.
`setfont`	`text-style(target, family, weight, posture, size)`	`native`	One command can set a complete text style for a target.
`featuresrm`, `featuressf`, `featurestt`, `featuresbf`, `featuresmd`, `featuresit`, `featuressl`, `featuressc`, `featuresup`, `featurestiny`, `featuresscriptsize`, `featuresfootnotesize`, `featuressmall`, `featuresnormalsize`, `featureslarge`, `featuresLarge`, `featuresLARGE`, `featureshuge`, `featuresHuge`	`typography(target: "features", ...)`	`native`	TeX's many short font macros are intentionally collapsed into target-based Typst typography controls.
`namesrm`, `numberingrm`, `residuesrm`, `rulerrm`, `legendrm`, `featurenamesrm`, `featurestylenames`, `featurestylenamesrm`, and corresponding family/shape/size variants	`typography(target: ..., ...)`, `text-family`, `text-weight`, `text-posture`, `text-size`	`native`	All target-specific TeX font shortcuts map to the same target-based text API.
Global document visual style	`theme: "screen"`, `theme: shade-theme(...)`, `theme: visual-theme(...)`	`native`	Themes are explicit values, not ambient macro state.
TeXshade preset-like command blocks	`preset: "publication"` or `shade-preset(name)`	`native`	Reusable presets are ordinary Typst values and can be imported from local files.

The legacy TeXshade typography shortcuts follow a regular naming scheme. Typshade maps each family to `typography(target: ..., ...)`, `text-family`, `text-weight`, `text-posture`, or `text-size`.

TeXshade shortcut family	**Covered suffixes**
`features*`	`rm`, `sf`, `tt`, `bf`, `md`, `it`, `sl`, `sc`, `up`, `tiny`, `scriptsize`, `footnotesize`, `small`, `normalsize`, `large`, `Large`, `LARGE`, `huge`, `Huge`
`featurestyles*`	`rm`, `sf`, `tt`, `bf`, `md`, `it`, `sl`, `sc`, `up`, `tiny`, `scriptsize`, `footnotesize`, `small`, `normalsize`, `large`, `Large`, `LARGE`, `huge`, `Huge`
`featurenames*`, `featurestylenames*`	`sf`, `tt`, `bf`, `md`, `it`, `sl`, `sc`, `up`, `tiny`, `scriptsize`, `footnotesize`, `small`, `normalsize`, `large`, `Large`, `LARGE`, `huge`, `Huge`
`names*`, `numbering*`, `residues*`, `legend*`	`sf`, `tt`, `bf`, `md`, `it`, `sl`, `sc`, `up`, `tiny`, `scriptsize`, `footnotesize`, `small`, `normalsize`, `large`, `Large`, `LARGE`, `huge`, `Huge`
`ruler*`, `rulername*`	`sf`, `tt`, `tiny`, `scriptsize`, `footnotesize`, `small`, `normalsize`, `large`, `Large`, `LARGE`, `huge`, `Huge`; `rulername*` also includes `rm`

14.11 Utility And Analysis Functions

TeXshade	**Typshade**	**Status**	**Migration Notes**
`molweight`	`molecular-weight(sequence, unit: "Da")`	`native`	Standalone sequence helper. Graph tracks can also use molecular-weight-derived metrics.
`charge`	`net-charge(sequence, termini: "o")`	`native`	Standalone sequence helper. Graph tracks can also use charge metrics.

`percentsimilarity`	`percent-similarity(source, sequence-a, sequence-b, ...)`	`native`	Computes pairwise percent similarity for an alignment source and optional displayed selection.
`percentidentity`	`percent-identity(source, sequence-a, sequence-b, ...)`	`native`	Computes pairwise percent identity over shared non-gap positions.
`similaritytable`, `identitytable`	`similarity-table(source, ...)`	`native`	Returns a Typst table with similarity values above the diagonal and identity values below the diagonal, matching TeXshade's combined table concept.
Current-alignment macro access	`alignment-data(source)`, `parse-alignment(text)`	`data`	The parsed alignment is explicit data that can be reused in Typst logic.
Sequence list diagnostics	`sequence-list(source)`	`data`	Document-facing sequence index/name table.
Selection diagnostics	`selection-preview(source, sequence, selection)`, `selection-table(source, ...)`	`data`	Preview motif/range/PDB selections without relying on TeX log output.

14.12 External Scripts And Side Effects

TeXshade includes helper behavior that prints messages, writes files, or generates external helper script text. Typshade deliberately avoids hidden render-time side effects:

TeXshade	Typshade	Status	Migration Notes
`exportconsensus`	Use `alignment-data` or an explicit build script.	`omitted`	No arbitrary file writes during Typst package evaluation.
`messagePDBlist`	`pdb-selection(...)` as content/data	`data`	The information is available, but it is not sent to TeX's console.
`printPDBlist`	`pdb-selection(...)` or `selection-table(...)`	`data`	Use document content rather than TeX output streams.
`structurememe`, `memeStandardcolors`, `memeRed`, `memeYellow`, `memeBlue`, `memeWhite`, `memeBlack`	External script generation outside the package.	`omitted`	Structure meme rendering requires generating Chimera-style command files; this side-effect belongs in an external build/tooling layer.
`chimerachain`, `chimeraballScale`, `memelabelcutoff`, `chimeraaxisdistance`, `chimeraxisdistance`, `echostructurefile`	External script generation outside the package.	`omitted`	These configure TeXshade's Chimera command-file output and are intentionally not public Typshade document-rendering controls.
Driver-dependent specials	Typst-native rendering only.	`omitted`	DVI/PDF driver hooks and TeX specials are not part of the Typst package model.

14.13 Complete Feature Coverage Summary

Typshade provides the TeXshade feature set through these current public groups:

Feature Family	Current Typshade Coverage
Alignment rendering	`shade`, `format:`, `seq-type:`, `commands:`, `figure:`, `regions:`, `features:`.
Purpose-level figures	`publication`, `motif-map`, `structure-map`, `logo-analysis`, `overview`.
Scoring and shading	`identical`, `similar`, `diverse`, `functional`, `single-sequence`, `tcoffee`, `scoring-mode`, `threshold`, `color-scheme`, `residue-style`, matrix/weight controls.
Sequence display	`lines`, `window`, `auto-layout`, `auto-page`, `fit:`, label and numbering controls, sequence visibility/order controls, gap controls, residue hiding, fingerprint controls.
Consensus and ruler tracks	`consensus`, `consensus-track`, `ruler`, `ruler-track`, symbols/colors/language/marker/rotation controls.
Logos and legends	`logo`, `sequence-logo`, `subfamily-logo`, frequency correction, negative values, relevance markers, logo colors, logo scale, legend controls.
Regions and annotations	`highlight`, `tint`, `emphasize`, `lower`, `frame`, `motif`, `mark`, `graph`, feature label and graph controls.
Translation features	`codon`, `genetic-code`, `backtranslation-label`, `backtranslation-text`, translation/complement feature styles.
Structure tracks	`structures`, `structure-tracks`, `dssp-track`, `stride-track`, `hmmtop-track`, `phd-topology-track`, `phd-secondary-track`, structure visibility and appearance controls.

Selections	`select`, `select-or`, `select-and`, `select-not`, `select-pad`, `select-all`, `select-range`, `select-residues`, `select-motif`, `select-metric`, plus PDB selection helpers.
PDB selections	`pdb-point`, `pdb-line`, `pdb-plane`, `pdb-selection`, `selection-preview`, `selection-table`.
Typography and layout	`typography`, `text-family`, `text-weight`, `text-posture`, `text-size`, `text-style`, spacing, separator, block-gap, line-gap, caption controls, `auto-layout`, `auto-page`.
Analysis utilities	`alignment-data`, `parse-alignment`, `alignment-summary`, `alignment-debug`, `cell-inspect`, `sequence-list`, `selection-preview`, `selection-table`, `percent-identity`, `percent-similarity`, `similarity-table`, `molecular-weight`, `net-charge`.

15 Public API Index

This index is organized from the Typshade side. Use it when you know the kind of object you want to control but do not remember the exact helper name.

Group	Public Names
Renderer	`shade`
Recipes	`publication`, `motif-map`, `structure-map`, `logo-analysis`, `overview`
Presets and themes	`shade-preset`, `shade-theme`, `visual-theme`, `resolve-color`, `scale-color`
Inspection and data	`alignment-data`, `parse-alignment`, `alignment-summary`, `alignment-debug`, `cell-inspect`, `selection-preview`, `sequence-list`, `selection-table`, `percent-identity`, `percent-similarity`, `similarity-table`
Selection DSL	`select`, `select-or`, `select-and`, `select-not`, `select-pad`, `select-all`, `select-range`, `select-residues`, `select-motif`, `select-metric`
Scoring shortcuts	`identical`, `similar`, `diverse`, `functional`, `single-sequence`, `tcoffee`
Layout shortcuts	`lines`, `window`, `names`, `no-names`, `numbers`, `no-numbers`, `no-numbering`, `typography`, `gap-style`
Track shortcuts	`consensus`, `no-consensus`, `ruler`, `no-ruler`, `logo`, `no-logo`, `legend`, `no-legend`, `structures`
Track controls	`consensus-track`, `ruler-track`, `ruler-marker`, `sequence-logo`, `no-sequence-logo`, `subfamily-logo`, `no-subfamily-logo`, `legend-track`, `structure-tracks`
Annotations	`highlight`, `tint`, `emphasize`, `mark`, `motif`, `graph`, `pdb-point`, `pdb-line`, `pdb-plane`
Core controls	`sequence-type`, `color-scheme`, `scoring-mode`, `tcoffee-scores`, `sequence-window`, `residues-per-line`, `auto-layout`, `auto-page`, `threshold`
Conservation controls	`shade-all-residues`, `all-match-threshold`, `disable-all-match-threshold`, `hide-all-match-positions`, `show-all-match-positions`, `weight-table`, `set-weight`, `gap-penalty`
Residue/group controls	`residue-style`, `cell-style`, `peptide-groups`, `dna-groups`, `peptide-similarities`, `dna-similarities`, `clear-functional-groups`, `functional-group`, `functional-style`
Names and numbering	`names-track`, `numbering-track`, `sequence-name`, `names-color`, `sequence-name-color`, `hide-sequence-name`, `numbering-color`, `sequence-number-color`, `hide-sequence-number`, `start-number`, `allow-zero-numbering`, `disallow-zero-numbering`, `sequence-length`
Consensus/ruler details	`consensus-name`, `consensus-language`, `consensus-symbols`, `consensus-colors`, `consensus-from-sequence`, `consensus-from-all-sequences`, `ruler-steps`, `ruler-color`, `ruler-name`, `ruler-name-color`, `ruler-space`, `rotate-ruler`, `unrotate-ruler`
Gap and sequence display	`gap-char`, `gap-rule`, `gap-colors`, `stop-char`, `show-leading-gaps`, `hide-leading-gaps`, `hide-residues`, `show-residues`, `keep-single-sequence-gaps`, `shift-single-sequence`
Regions and sequence rows	`domain`, `domain-gap-rule`, `domain-gap-colors`, `highlight-block`, `region-color-scheme`, `lower`, `lower-block`, `emphasis-block`, `tint-block`, `tint-default`, `emphasis-default`, `frame`, `hide-sequence`, `hide-all-sequences`, `show-all-sequences`, `remove-sequence`, `no-shade`, `separation-line`, `sequence-order`
Feature and translation controls	`feature-rule`, `codon`, `genetic-code`, `backtranslation-label`, `backtranslation-text`, `feature-text-label`, `feature-style-label`, `hide-feature-text-label`, `hide-feature-style-label`, `hide-feature-text-labels`, `hide-feature-style-labels`, `feature-text-label-color`, `feature-style-label-color`, `feature-text-label-color-at`, `feature-style-label-color-at`
Logo controls	`frequency-correction`, `no-frequency-correction`, `subfamily`, `sequence-logo-name`, `subfamily-logo-name`, `logo-scale`, `no-logo-scale`, `logo-stretch`, `negative-logo-values`, `no-negative-logo-values`, `relevance-threshold`, `relevance-marker`, `no-relevance-marker`, `logo-color`, `clear-logo-colors`
Legend controls	`legend-color`, `legend-offset`, `color-swatch`

Structure controls	`show-structure-types`, `hide-structure-types`, `structure-appearance`, `use-first-dssp-column`, `use-second-dssp-column`, `stride-track`, `dssp-track`, `hmmtop-track`, `phd-topology-track`, `phd-secondary-track`
Graph/layout controls	`bar-graph-stretch`, `color-scale-stretch`, `alignment-position`, `character-stretch`, `line-stretch`, `numbering-width`, `fingerprint`, `align-right-labels`, `align-left-labels`, `auto-layout`, `auto-page`. Default block placement is left-aligned.
Typography and spacing	`text-family`, `text-weight`, `text-posture`, `text-size`, `text-style`, `caption`, `short-caption`, `small-separator`, `medium-separator`, `large-separator`, `no-block-gap`, `small-block-gap`, `medium-block-gap`, `large-block-gap`, `block-gap`, `flexible-block-gap`, `fixed-block-gap`, `no-line-gap`, `small-line-gap`, `medium-line-gap`, `large-line-gap`, `line-gap`, `feature-slot-space`
Sequence utilities	`molecular-weight`, `net-charge`, `pdb-selection`

16 Public API Example Catalog

This catalog gives an example invocation for every public Typshade function and pairs each feature family with a compiled Typst result. The snippets are intentionally compact, but the result panels are real document content produced by the same package during compilation. When several low-level controls belong to one feature family, the sample code lists the individual controls and the typeset result demonstrates the family in a compact alignment or table.

16.1 Rendering, Recipes, Presets, Themes, And Colors

Use `shade` for rendering, recipe functions for common figure purposes, presets and themes for reusable visual defaults, and color helpers when building custom Typst logic around Typshade colors.

```
#let alignment = read("alignment.msf", encoding: none)

#shade(alignment, format: "msf")

#shade(alignment, format: "msf", figure: publication(region: "80..112"))
#shade(alignment, format: "msf", figure: motif-map(("NPA": "loop")))
#shade(alignment, format: "msf", figure: structure-map(1, topology: read("AQP1.top", encoding:
none)))
#shade(alignment, format: "msf", figure: logo-analysis(sequence: 3, region: "203..235"))
#shade(alignment, format: "msf", figure: overview(colors: "grays"))
#let resolved-figure = resolve-figure(
  alignment,
  "msf",
  publication(region: "80..112"),
)

#shade(alignment, format: "msf", preset: "publication")
#shade(alignment, format: "msf", theme: "screen")
#shade(
  alignment,
  format: "msf",
  theme: visual-theme(names: "RoyalBlue", ruler: "DarkGray"),
)

#let preset-commands = shade-preset("publication")
#let theme-commands = shade-theme((names: "RoyalBlue", ruler: "DarkGray"))
#let resolved-red = resolve-color("BrickRed")
#let midpoint = scale-color("ColdHot", 50)
```

Typeset result Rendering, recipes, presets, themes, and colors

Basic **`shade`**

```
Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2
consensus !  *
```

`publication(...)`

```
          80        90        100       110
          |         |         |         |
AQP1.PRO  TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO  TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO  TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO  TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO  TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
consensus ! * *    ** *****!  !! *!!* ***!*
```

motif-map(...)

```
          80        90        100       110
          |         |         |         |
AQP1.PRO  TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO  TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO  TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO  TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO  TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
consensus ! * *    ** *****!  !! *!!* ***!*
```

structure-map(...)

```
                     10        20        30        40
                     |         |         |         |
AQP1.PRO    1 MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLERNQTL 45
AQP2.PRO    1 MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...SS.... 37
AQP3.PRO    1 M........LLRQALAECLGTLILVMFGCGSVAQVVLSRGTH... 46
AQP4.PRO    1 MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...GSENPL 66
AQP5.PRO    1 MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...SA.... 38
consensus     !  *    ** *** !!*!*****!*** !!** *
```

logo-analysis(...)


```
AQP1.PRO  TGCGINPARSFGSAVLTR..NFS.N......HW 210
AQP2.PRO  TGCSMNPARSLAPAVVTG..KFD.D......HW 202
AQP3.PRO  SGYAVNPARDFGPRLFTALAGWGSEVFTTGQNW 235
AQP4.PRO  TGASMNPARSFGPAVIMG..NWE.N......HW 231
AQP5.PRO  TGCSMNPARSFGPAVVMN..RFSPS......HW 204
consensus *!***!!!!****** *    *         *!
```

overview(...)

```
AQP1.PRO MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLERNQTLVQDNV
AQP2.PRO MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...SS....PPSVL
AQP3.PRO M........LLRQALAECLGTLILVMFGCGSVAQVVLSRGTH....GGFL
AQP4.PRO MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...GSENPLPVDMV
AQP5.PRO MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...SA....LPTIL
```

Preset and theme

```
Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2
consensus ! *
```

Data helpers for recipes and colors

`resolve-figure`: 15 command fragments
`shade-preset`: 5 fragments
`shade-theme`: 3 fragments
`resolve-color`:
`scale-color`:

16.2 Data, Inspection, And Analysis

These functions return parsed alignment data, diagnostic tables, or numerical sequence comparisons. They are useful both in prose and in figure-building logic.

```
#let alignment = read("alignment.msf", encoding: none)
#let parsed = alignment-data(alignment, format: "msf")
#let tiny = parse-alignment(">A\nACD\n>B\nA-D", format: "fasta")

#alignment-summary(alignment, format: "msf")
#sequence-list(alignment, format: "msf")
#alignment-debug(alignment, format: "msf", commands: (lines(auto),))
#selection-preview(alignment, 1, "80..112", format: "msf")
#selection-table(
  alignment,
  "80..112",
  "NPA",
  format: "msf",
  sequence: 1,
)
#cell-inspect(alignment, 1, 80, format: "msf")

#percent-identity(alignment, 1, 2, format: "msf")
#percent-similarity(alignment, 1, 2, format: "msf")
#similarity-table(alignment, format: "msf")
```

Typeset result Data, inspection, and analysis helpers

`alignment-data` and `parse-alignment`

Parsed fixture: 4 sequences / 4 columns
Parsed inline: 2 sequences / 3 columns

`alignment-summary` and `sequence-list`

Field	Value
Format	FASTA
Type	P
Sequences	4
Columns	4
Names	Alpha, Beta, Gamma, seq4

Name	Length	Non-gap residues
Alpha	4	3
Beta	4	3
Gamma	4	2
seq4	4	2

`alignment-debug`

Field	Value
Format	FASTA
Type	P
Sequences	4
Visible sequences	Alpha, Beta, Gamma, seq4

Columns	4
Displayed columns	4
Residues per line	auto
Scoring mode	similar
Consensus source	all

`selection-preview` and `selection-table`

Motif columns: 1,2

Name	Selection	Positions	Count
motif	A[ED]	1,2	2
first two	1..2	1,2	2

`cell-inspect`

Field	Value
Sequence	Alpha
Column	1
Position	1
Residue	A
Kind	allmatch
Rendered char	A
Foreground	Goldenrod
Background	RoyalPurple
Consensus	A
Consensus score	100

Pairwise analysis

Identity 1/2: 66.7%
Similarity 1/2: 100%

	Alpha	Beta	Gamma	seq4
Alpha	-	100	50	50
Beta	66.7	-	50	50
Gamma	50	50	-	100
seq4	50	50	100	-

16.3 Auto Layout, Pagination, And Selection DSL

Auto layout uses Typst layout information to choose the rendered line length. Selection DSL helpers produce selection values that can be passed anywhere a range, motif, or PDB selection is accepted.

```
#let alignment = read("alignment.msf", encoding: none)

#shade(
  alignment,
  format: "msf",
  fit: "container",
  commands: (
    window(1, select-or(
      select-range(80, 112),
      select-motif("NPA"),
      select-metric("coverage", at-least: 80),
    )),
  ),
)

#shade(
  alignment,
  format: "msf",
  fit: (mode: "page", min: 25, max: 55, page: (blocks: auto)),
```

```
  commands: (
    auto-layout(max: 55),
    auto-page(blocks: auto, repeat-legend: true),
    highlight(1, select-and(select-range(80, 112), select-not(select-residues(95))), bg:
"LightYellow"),
    mark("top", 1, select-pad(select-residues(93), 2), text: "site"),
  ),
)

#selection-table(
  alignment,
  (name: "all", selection: select-all()),
  (name: "motif", selection: select-motif("NPA")),
  (name: "combined", selection: select(select-range(80, 112), select-metric("coverage", at-
least: 80))),
  format: "msf",
)
```

Typeset result Auto layout, pagination, and Selection DSL

Selection DSL previews

Name	Selection	Positions	Count
\`select-all\`	all	1,2,3	3
\`select-range\`	range	1,2	2
\`select-residues\`	positions	1,3	2
\`select-motif\`	motif	1,2	2
\`select-metric\`	metric	1,2,3	3
\`select-or\`	or	1,3	2
\`select-and\`	and	1,2	2
\`select-not\`	not	1,3	2
\`select-pad\`	pad	1,2,3	3

Fit and auto commands

`auto-layout(max: 4)`	**auto-layout**
`auto-page(blocks: 1)`	auto-page

```
Alpha AEF3
Beta  ADF3
Gamma A.G2
seq4  AC.2
```

16.4 Scoring, Shading, And Sequence-Type Controls

Scoring helpers are usually placed in `commands:`. The lower-level controls are available when a figure needs exact threshold, residue style, similarity, or weight-table behavior.

```
#let alignment = read("alignment.msf", encoding: none)
#let tcoffee-data = read("AQP_TC.asc", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    sequence-type("P"),
    identical(colors: "blues", threshold: 50),
    similar(colors: "greens", threshold: 45),
    diverse(option: 1),
    functional("charge"),
    single-sequence(sequence: 1),
    tcoffee(tcoffee-data),
    scoring-mode("similar"),
    tcoffee-scores(tcoffee-data),
```

```
    color-scheme("reds"),
    threshold(60),
    all-match-threshold(value: 90),
    disable-all-match-threshold(),
    shade-all-residues(),
    hide-all-match-positions(),
    show-all-match-positions(),
    weight-table("BLOSUM62"),
    set-weight("E", "Q", 2),
    gap-penalty(-5),
  ),
)
```

```
#shade(
  alignment,
  format: "msf",
  commands: (
    residue-style("conserved", "White", "RoyalBlue", case: "upper", style: "bold"),
    peptide-groups(("FYW", "ILVM", "RK", "DE", "GA", "ST", "NQ")),
    dna-groups(("GAR", "CTY")),
    peptide-similarities("S", "TA"),
    dna-similarities("A", "GR"),
    clear-functional-groups(),
    functional-group("acidic (-)", "DE", "White", "Red"),
    functional-style("D", "White", "BrickRed"),
  ),
)
```

Typeset result Scoring, shading, residue styles, and sequence type controls

`similar`, `threshold`, and color scheme

```
AQP1.PRO  GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO  AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO  GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO  GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO  AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
consensus *   ****  !  *!*********!  **  *   !*  **
```

`identical`, custom residue style, and `cell-style`

```
Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2
consensus !  *
```

`functional`, groups, and legend

```
acidic
basic

AQP1.PRO  GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO  AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO  GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO  GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO  AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
consensus ********!**!*********!*******!****
```

`diverse`, weights, and gap penalty

```
AQP1.PRO . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 170
AQP2.PRO . . t . . . f l . . . . . . . . f . . . . e . . g . n l . . p a . 162
AQP3.PRO . f f d . f . . . a a . . . . . . . i v . p n n p v p r . l e a f 186
AQP4.PRO . . l . . . . i . f . . . f t . f . . c . s . . t . . t . . v a . 191
AQP5.PRO . . v . . . . l . f . . a . . . f . . . . s . . t s p v . . p a . 163
```

single-sequence

```
              140            150            160            170
               |              |              |              |
AQP1.PRO GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
```

tcoffee and T-Coffee scale

```
AQP1.PRO     GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO     AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO     GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO     GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO     AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
reliability  *    ****  !  *!*********!  **  *    !*  **
```

16.5 Windows, Labels, Numbering, Rulers, And Gaps

The shortcut functions cover common cases. The track/control functions expose individual knobs for sequence labels, numbering, ruler names, ruler markers, gap characters, stop symbols, and leading-gap behavior.

```
#shade(
  alignment,
  format: "msf",
  commands: (
    lines(40),
    residues-per-line(40),
    window(1, "80..112"),
    sequence-window(1, "80..112", start: 80),

    names(position: "left", color: "RoyalBlue"),
    no-names(),
    numbers(position: "right", color: "DarkGray"),
    no-numbers(),
    names-track(position: "left", color: "RoyalBlue"),
    no-numbering(),
    numbering-track(position: "leftright", color: "DarkGray"),

    sequence-name(1, "AQP1"),
    names-color("RoyalBlue"),
    sequence-name-color((1, 3), "Red"),
    hide-sequence-name((2,)),
    numbering-color("DarkGray"),
    sequence-number-color((1, 3), "Red"),
    hide-sequence-number((2,)),

    start-number(1, 80, selection: "80..112"),
    allow-zero-numbering(),
    disallow-zero-numbering(),
    sequence-length(1, 269),
  ),
)
```

```
#shade(
  alignment,
  format: "msf",
  commands: (
    ruler("top", sequence: 1, every: 10, name: "AQP1 positions"),
    no-ruler("bottom"),
    ruler-track(position: "bottom", sequence: 1, steps: 5, color: "DarkGray"),
```

```
    ruler-marker(100, "site", position: "top", color: "Red"),
    ruler-steps(10),
    ruler-color("DarkGray"),
    ruler-name("positions"),
    ruler-name-color("Red"),
    ruler-space(2pt),
    rotate-ruler(),
    unrotate-ruler(),

    gap-style(foreground: "Gray60", background: "White", rule: 0.5pt),
    gap-char("."),
    gap-rule(0.5pt),
    gap-colors("Gray60", "White"),
    stop-char("*"),
    show-leading-gaps(),
    hide-leading-gaps(),
  ),
)
```

Typeset result Windows, labels, numbering, rulers, and gaps

Window, names, and numbering

```
AQP1       80 TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO   72 TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO   80 TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO  101 TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO   73 TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
```

Hidden labels and numbering

```
    1 AEF. 3    Alpha
      ADF.
    1 A.G. 2    Gamma
    1 AC.. 2     seq4
```

Ruler details

```
AQP1.PRO    TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO    TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO    TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO    TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO    TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
            |    |    |    |    |    |    |
positions   80   85   90   95   100  105  110
```

Gap and stop controls

```
Alpha -AE!3
Beta  -A-!2
```

`no-names`, `no-numbering`, and hidden leading gaps

```
AE*
A *
```

16.6 Consensus, Logos, Legends, And Swatches

Consensus, logo, and legend helpers can be used as shortcuts or as explicit tracks. Logo controls cover frequency correction, subfamily views, scale labels, negative values, relevance markers, and custom residue colors.

```
#shade(
  alignment,
  format: "msf",
  commands: (
    consensus("bottom", scale: "ColdHot", name: "conservation"),
    no-consensus(),
    consensus-track(position: "top", scale: "BlueRed", name: "consensus"),
    consensus-name("agreement"),
    consensus-language("english"),
    consensus-symbols(".", "lower", "upper"),
    consensus-colors(
      none-fg: "Black",
      none-bg: "White",
      conserved-fg: "White",
      conserved-bg: "RoyalBlue",
      allmatch-fg: "Yellow",
      allmatch-bg: "RoyalPurple",
    ),
    consensus-from-sequence(1),
    consensus-from-all-sequences(),
  ),
)
```

```
#shade(
  alignment,
  format: "msf",
  commands: (
    logo("top", colors: "charge", name: "logo"),
    no-logo(),
    sequence-logo(position: "top", colors: "rasmol", name: "logo"),
    no-sequence-logo(),
    subfamily-logo((3,), position: "bottom", colors: "charge", name: "AQP3", negative-name:
"others"),
    no-subfamily-logo(),
    frequency-correction(),
    no-frequency-correction(),
    subfamily((3,)),
    sequence-logo-name("sequence logo"),
    subfamily-logo-name("AQP3", negative-name: "others"),
    logo-scale(position: "leftright", color: "DarkGray"),
    no-logo-scale(),
    logo-stretch(1.2),
    negative-logo-values(),
    no-negative-logo-values(),
    relevance-threshold(1.0),
    relevance-marker(char: "*", color: "Red"),
    no-relevance-marker(),
    logo-color("DE", "Red"),
    clear-logo-colors(default: "Black"),
  ),
)

#shade(alignment, format: "msf", commands: (legend(color: "Black"),))
#shade(alignment, format: "msf", commands: (no-legend(), legend-track(color: "Black")))
#shade(alignment, format: "msf", commands: (legend-color("DarkGray"), legend-offset(6pt, 0pt)))
#color-swatch("RoyalBlue")
```

Typeset result Consensus, logos, legends, and swatches

Consensus symbols and colors

```
Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2
consensus Aef.
```

All-sequence consensus and no consensus

```
consensus !  *
Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2

Alpha AEF.3
Beta  ADF.3
Gamma A.G.2
seq4  AC..2
```

Sequence logo controls


```
AQP1.PRO TGCGINPARSFGSAVLTR..NFS.N......HW210
AQP2.PRO TGCSMNPARSLAPAVVTG..KFD.D......HW202
AQP3.PRO SGYAVNPARDFGPRLFTALAGWGSEVFTTGQNW235
AQP4.PRO TGASMNPARSFGPAVIMG..NWE.N......HW231
AQP5.PRO TGCSMNPARSFGPAVVMN..RFSPS......HW204
```

Subfamily logo controls

```
AQP1.PRO TGCGINPARSFGSAVLTR..NFS.N......HW210
AQP2.PRO TGCSMNPARSLAPAVVTG..KFD.D......HW202
AQP3.PRO SGYAVNPARDFGPRLFTALAGWGSEVFTTGQNW235
AQP4.PRO TGASMNPARSFGPAVIMG..NWE.N......HW231
AQP5.PRO TGCSMNPARSFGPAVVMN..RFSPS......HW204
```

Logo/legend toggles

```
Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2
consensus !  *
```

Legend and swatches

```
acidic (-)
basic (+)

Alpha     AEF.3
Beta      ADF.3
Gamma     A.G.2
seq4      AC..2
consensus !**
```

16.7 Regions, Domains, Motifs, Marks, Graphs, And PDB Selections

Region and annotation helpers handle the common TeXshade feature patterns: selected domains, block highlighting, tinting, emphasis, lowercase regions, frames, marks, motifs, graph tracks, and structure-derived selections.

```
#let pdb = read("1J4N.pdb", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    domain(1, "80..90,100..112"),
    domain-gap-rule(0.6pt),
    domain-gap-colors("Gray70", "White"),

    highlight(1, "NPA", fg: "White", bg: "BrickRed"),
    highlight-block(1, "80..112", "White", "LightBlue"),
    region-color-scheme(1, "80..112", "reds"),
    tint(1, "158..163", intensity: "strong"),
    tint-block(1, "158..163", intensity: "weak"),
    tint-default("normal"),
    emphasize(1, "NPA", style: "italic"),
    emphasis-block(1, "158..163", style: "bold"),
    emphasis-default("italic"),
    lower(1, "80..90"),
    lower-block(1, "80..90"),
    frame(1, "82..82,106..106", color: "Red"),
  ),
)
```

```
#shade(
  alignment,
  format: "msf",
  commands: (
    mark("top", 1, "93..93", fill: "Yellow", text: "site"),
    motif(1, "NXX[ST]N", text: "glycosylation", position: "top"),
    graph("top", 1, "138..170", "conservation", kind: "bar"),
    graph("top", 1, "138..170", "entropy", kind: "color", options: ("WhiteBlack",)),
    graph("bottom", 1, "138..170", "coverage", kind: "bar"),
    graph("bottom", 1, "138..170", "hydrophobicity", kind: "color", options: ("ColdHot",)),
    bar-graph-stretch(1.4),
    color-scale-stretch(1.2),
    feature-rule(0.7pt),
    feature-text-label("top", "features"),
    feature-style-label("bottom", "graph"),
    hide-feature-text-label("top"),
    hide-feature-style-label("bottom"),
    hide-feature-text-labels(),
    hide-feature-style-labels(),
    feature-text-label-color("DarkGray"),
    feature-style-label-color("DarkGray"),
    feature-text-label-color-at("top", "Red"),
    feature-style-label-color-at("bottom", "Blue"),
  ),
)
```

```
#let point = pdb-point(pdb, 81, distance: 8, atom: "CA")
#let line = pdb-line(pdb, 81, 168, distance: 1)
#let plane = pdb-plane(pdb, 66, 73, 199, distance: 1)

#shade(
  alignment,
  format: "msf",
  commands: (
    window(1, point),
    highlight(1, line, bg: "Yellow"),
    mark("top", 1, plane, text: "plane"),
  ),
)

#pdb-selection(point)
```

Typeset result Regions, motifs, marks, graphs, and PDB selections

Domains and domain gaps

```
         80            90100         110
         |             | |           |
AQP1.PRO TLGLLLSCQISAQCVGAIVASAIL 112
AQP2.PRO TVACLVGCHVSAQLLGAVAGAAIL 104
AQP3.PRO TFAMCFLAREPAQTLGAFLGAGIV 112
AQP4.PRO TVAMVCTRKISAQCLGAIIGAGIL 133
AQP5.PRO TLALLIGNQISAQLVGAIAGAGIL 105
```

Highlights, tint, emphasis, lower, and frame

```
AQP1.PRO glGIEIIGTLQLVLCVLATTDRRRRDLGGSApl 170
AQP2.PRO AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
```

Marks and motifs

```
AQP1.PRO TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
```

Graph metrics and graph controls

```
AQP1.PRO GLGIEIIGTLQLVLCVLATTDRRRRDLGGSAPL 170
AQP2.PRO AVTVELFLTMQLVLCIFASTDERRGDNLGSPAL 162
AQP3.PRO GFFDQFIGTAALIVCVLAIVDPNNPVPRGLEAF 186
AQP4.PRO GLLVELIITFQLVFTIFASCDSKRTDVTGSVAL 191
AQP5.PRO AMVVELILTFQLALCIFSSTDSRRTSPVGSPAL 163
```

Feature label hiding

```
AQP1.PRO TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
```

PDB selections

```
      123
      |||
Alpha AEF. 3
Beta  ADF. 3
Gamma A.G. 2
seq4  AC.. 2
```

16.8 Sequence Visibility, Ordering, Layout, Typography, And Spacing

These controls cover sequence visibility, row ordering, separators, figure placement, character metrics, fingerprints, label alignment, target-specific typography, captions, and block/line/feature spacing.

```
#shade(
  alignment,
  format: "msf",
  commands: (
    hide-sequence(2),
    hide-all-sequences(),
    show-all-sequences(),
    remove-sequence(5),
    no-shade((1, 3)),
```

```
    separation-line(3),
    sequence-order((5, 4, 3, 2, 1)),

    hide-residues(),
    show-residues(),
    fingerprint(360),
    alignment-position("left"),
    character-stretch(1.1),
    line-stretch(1.05),
    numbering-width(6),
    align-right-labels(),
    align-left-labels(),
  ),
)
```

```
#shade(
  alignment,
  format: "msf",
  commands: (
    typography(target: "all", family: "mono", size: 8pt),
    text-family("names", "mono"),
    text-weight("names", "bold"),
    text-posture("features", "italic"),
    text-size("residues", 7pt),
    text-style("ruler", "sans", "regular", "normal", 7pt),

    caption([Aquaporin alignment], position: "bottom"),
    short-caption([AQP alignment]),
    small-separator(),
    medium-separator(),
    large-separator(),
    no-block-gap(),
    small-block-gap(),
    medium-block-gap(),
    large-block-gap(),
    block-gap(0.8em),
    flexible-block-gap(),
    fixed-block-gap(),
    no-line-gap(),
    small-line-gap(),
    medium-line-gap(),
    large-line-gap(),
    line-gap(0.2pt),
    feature-slot-space("top", 2pt),
  ),
)
```

Typeset result Sequence visibility, ordering, layout, typography, and spacing

Visibility, ordering, separators, and no-shade

```
Beta  ADF.3

first AEF.3
```

Hidden residues, fingerprint, and alignment

```
Alpha
 Beta
Gamma
 seq4
```

```
Alpha AEF.3
Beta  ADF.3
Gamma A.G.2
seq4  AC..2
```

Typography controls

```
           80        90        100       110
           |         |         |         |
AQP1.PRO TLGLLLSCQISILRAVMYIIAQCVGAIVASAIL 112
AQP2.PRO TVACLVGCHVSFLRAAFYVAAQLLGAVAGAAIL 104
AQP3.PRO TFAMCFLAREPWIKLPIYTLAQTLGAFLGAGIV 112
AQP4.PRO TVAMVCTRKISIAKSVFYITAQCLGAIIGAGIL 133
AQP5.PRO TLALLIGNQISLLRAVFYVAAQLVGAIAGAGIL 105
```

Captions and spacing commands

```
Top caption
Alpha AE 2
Beta  AD 2
Gamma A. 1
seq4  AC 2

Alpha F. 3
Beta  F. 3
Gamma G. 2
seq4  ..
Bottom caption
```

Block and line gap presets

```
Alpha AE 2
Beta  AD 2
Gamma A. 1
seq4  AC 2

Alpha F. 3
Beta  F. 3
Gamma G. 2
seq4  ..

Alpha AE 2

Beta  AD 2

Gamma A. 1

seq4  AC 2


Alpha F. 3

Beta  F. 3

Gamma G. 2

seq4  ..
```

Native Typst figure wrapper

```
Alpha AEF.3
Beta  ADF.3
Gamma A.G.2
seq4  AC..2
```

Figure 1: Alignment wrapped by Typst `figure`

16.9 Translation, Structure Tracks, And Sequence Utilities

Translation helpers configure codon handling and backtranslation display. Structure helpers attach DSSP, STRIDE, HMMTOP, PHD topology, or PHD secondary structure data and control which structure classes are visible.

```
#let dna = read("dna-alignment.msf", encoding: none)

#shade(
  dna,
  format: "msf",
  seq-type: "N",
  commands: (
    single-sequence(sequence: 1),
    keep-single-sequence-gaps(),
    shift-single-sequence(value: -1),
    codon("A", "GCA,GCG,GCC,GCT,GCU,GCN"),
    genetic-code("standard"),
    backtranslation-label(style: "oblique", size: "tiny"),
    backtranslation-text(style: "horizontal", size: "tiny"),
    mark("bottom", 1, "1..90", style: "translate[Red]", text: "translation"),
    mark("bottom", 1, "1..90", style: "complement[LightBlue][lower]", text: "complement"),
  ),
)
```

```
#let topology = read("AQP1.top", encoding: none)
#let secondary = read("AQP1.phd", encoding: none)
#let hmmtop = read("AQP_HMM.ext", encoding: none)
#let dssp = read("model.dssp", encoding: none)
#let stride = read("model.stride", encoding: none)

#shade(
  alignment,
  format: "msf",
  commands: (
    structures(
      1,
      topology: topology,
      secondary: secondary,
      hmmtop: hmmtop,
    ),
    structure-tracks(
      1,
      topology: topology,
      secondary: secondary,
      hmmtop: hmmtop,
    ),
    phd-topology-track(1, topology),
    phd-secondary-track(1, secondary),
    hmmtop-track(1, hmmtop, source-sequence: 1),
    dssp-track(1, dssp),
    stride-track(1, stride),
    show-structure-types(
      "PHDtopo",
      ("TM", "internal", "external"),
    ),
    hide-structure-types("PHDtopo", ("internal",)),
    structure-appearance(
      "PHDtopo",
      "TM",
      "top",
      "box[Blue]:TM",
      "",
    ),
    use-first-dssp-column(),
    use-second-dssp-column(),
  ),
)
```

```
#molecular-weight("ACDEFGHIK", unit: "Da")
#molecular-weight("ACDEFGHIK", unit: "kDa")
#net-charge("ACDEFGHIK", termini: "o")
#net-charge("ACDEFGHIK", termini: "i")
```

Typeset result Translation, structure tracks, and sequence utilities

Translation and complement feature styles

```
AQP1nuc.SEQ CCTGGGCATTGAGATCATTGGCACCCTGCA 443
```

Structure tracks and appearance

```
AQP1.PRO MASEIKKKLFWRAVVAEFLAMTLFVFISIGSALGFNYPLERNQTLVQDNV 50
AQP2.PRO MW.ELRSIAFSRAVLAEFLATLLFVFFGLGSALQWA...SS....PPSVL 42
AQP3.PRO M........LLRQALAECLGTLILVMFGCGSVAQVVLSRGTH....GGFL 50
AQP4.PRO MSDGVWTQAFWKAVTAEFLAMLIFVLLSVGSTINWG...GSENPLPVDMV 71
AQP5.PRO MK.EVCSLAFFKAVFAEFLATLIFVFFGLGSALKWP...SA....LPTIL 43
```

DSSP/STRIDE command constructors

`dssp-track`	**dssp-track**
`stride-track`	stride-track
`use-first-dssp-column`	use-first-dssp-column
`use-second-dssp-column`	use-second-dssp-column

Sequence utilities

`molecular-weight("ACDEFGHIK")`	**1163 Da**
`molecular-weight(..., "kDa")`	1.16 kDa
`net-charge("ACDEFGHIK")`	−0.96
`net-charge(..., termini: "i")`	−0.87

17 Typst-Specific Improvements

- `figure:` recipes let hand-written figures read as scientific intent.
- `commands:` keeps custom figures explicit without adding another top-level API.
- Dictionaries and arrays make presets easy to share.
- Inspection helpers such as `alignment-summary`, `sequence-list`, and `selection-table` can be embedded in documents.
- `alignment-data` and `parse-alignment` expose parsed data for custom Typst logic.
- HTML export uses Typst's HTML target to produce scrollable SVG alignment frames and semantic captions.
- Public names follow Typst conventions instead of TeX macro naming.

17.1 HTML Export

Typst HTML export is experimental and currently requires the HTML feature flag. Compile with `--features html --format html`; Typshade will detect the HTML target and wrap `shade(...)` output in a scrollable `html.frame` SVG. A `caption:` on `shade(...)` becomes semantic `figure`/`figcaption` HTML. Data-report helpers such as `selection-table` emit native HTML tables with explicit border styling.

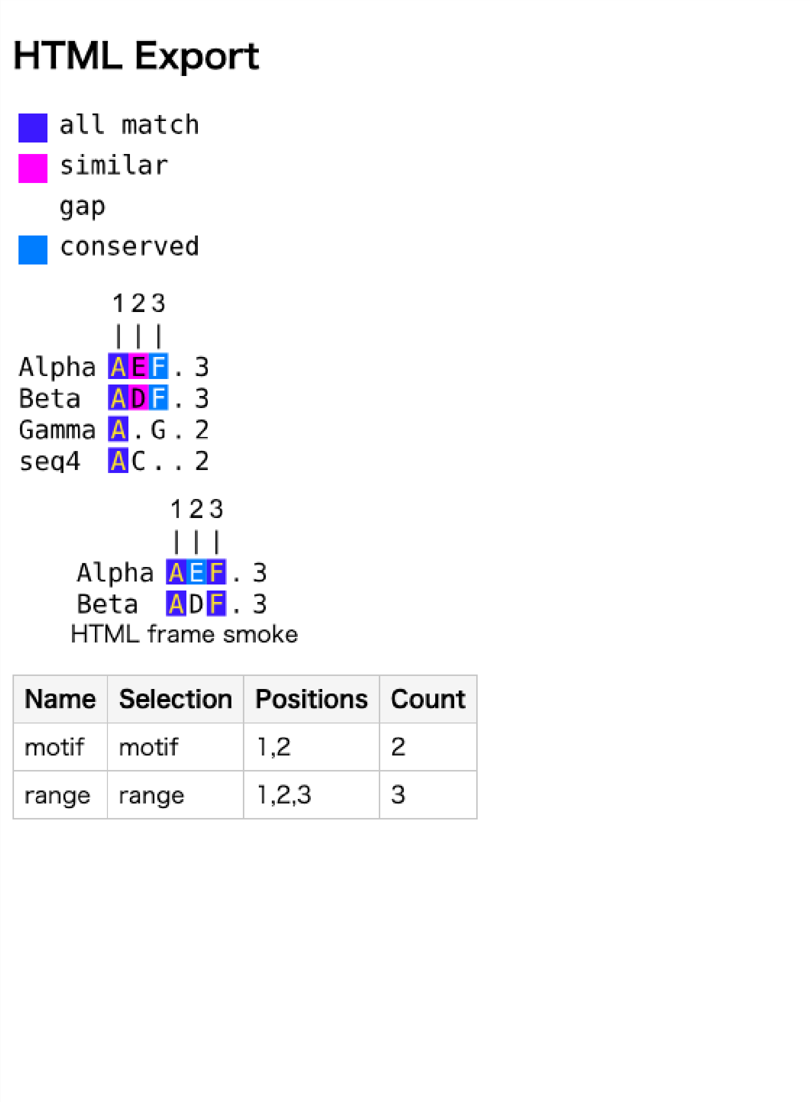
The following smoke-test source is exported with the command below. The browser screenshot is generated from the same source.

```
typst compile --features html --format html --root . tests/html-export.typ html-export.html
```

```
//! Smoke fixture for semantic Typst HTML export.

#import "../package/lib.typ": *

#set text(size: 8pt)

#let protein = read("fixtures/tiny-protein.fasta", encoding: none)
#let msf = read("fixtures/tiny.msf", encoding: none)

= HTML Export
```

```
#shade(
  protein,
  format: "fasta",
  residues-per-line: 4,
  legend: true,
  commands: (
    similar(colors: "blues", threshold: 50),
    ruler("top", sequence: 1, every: 1),
    no-consensus(),
  ),
)

#shade(
  msf,
  format: "msf",
  caption: [HTML frame smoke],
  fit: "container",
  commands: (
    identical(threshold: 50),
    ruler("top", sequence: 1, every: 1),
    no-consensus(),
  ),
)

#selection-table(
  protein,
  (name: "motif", selection: select-motif("A[ED]")),
  (name: "range", selection: select-range(1, 3)),
  format: "fasta",
)
```

Browser-rendered HTML Chrome rendering of the exported HTML



18 Known Typst Differences

Some TeXshade behavior depends on TeX box construction, console messages, or external file writes. Typshade keeps equivalent document-facing behavior where it is meaningful, but does not promise byte-for-byte or box-for-box output.

Area	Typshade behavior
External file writes	Package evaluation does not perform arbitrary side-effect exports. Return data/content or use external tooling when needed.
Console messages	Inspection helpers return Typst content/data instead of writing TeX messages.
Page breaking glue	Typst uses deterministic layout controls rather than TeX glue side effects.
Exact box geometry	Ruler and backtranslation geometry is represented in Typst layout, not TeX box primitives.

19 License

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